EIGSEP Memo 002

Marjum Pass 2026-07: images, camera poses and positions

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Abstract

Where the antenna and transmitter were, how we know, and how well. The positions come from photogrammetry of 37 field photographs against a digital elevation model (DEM); the absolute azimuth reference comes from GPS points on the highline anchors. The adopted geometry is `v0005`: a joint MCMC fit of all 29 camera poses, 3,993 landmarks, antenna and transmitter in the EPSG:6341 UTM raster grid of DEM `v0002`. It replaces the provisional point fit `v0004`, adding multi-view tie points that link every camera into one network. Its chains have not converged: 210 of the 213 global coordinates fail $\hat{R} \leq 1.01$ (median 1.05, max 1.36) and the smallest bulk effective sample size is 11.8. For the antenna and transmitter coordinates $\hat{R}$ is 1.07–1.14 and the five chain means differ by up to 0.24 m, so their posterior intervals are approximate. Refits withholding one kind of evidence at a time bound the antenna to 3.5 m and the transmitter to 0.1 m. This memo states the adopted positions, those checks, how they compare with the historical release `v0001` and provisional `v0004`, what the joint MCMC establishes, and the camera and coordinate conventions needed to reuse any of it. Every figure and table is generated by `make_products.py` in this directory (Appendix D). The observing configuration and angle conventions are in memo 001.

1 Summary

Table 1 gathers the adopted coordinates and orientations alongside their marginalized 95% credible intervals as key findings of the campaign geometry.

Table 1: Key campaign geometry parameters: adopted values and marginalized 95% credible intervals. The `v0005` chains have not converged, so their intervals are approximate (§5). Adopted positions and feed axes are posterior means over the `v0005` chain draws (§5, Fig. 7). The ground-antenna position and dipole bearing come from the `box-gnd` fit (§9.3), not the chains. Coordinates are UTM grid (DEM `v0002`); UTM easting/northing add the raster origin $(291000.25, 4345000.25)$ m. True bearings differ from grid bearings by the UTM grid convergence ($\gamma = +1.52°$ at the highline; true = grid $- 1.52°$).

Target / Quantity	Component	Frame / Units	Adopted	Marginalized 95% Interval	Constraint / Method
Sky antenna (91 m)	E (Easting)	UTM grid [m]	1655.13	[1654.64, 1655.68]	HMC joint posterior (v0005)
	N (Northing)	UTM grid [m]	2030.96	[2030.68, 2031.25]	95% 3D radius: 0.59 m
	U (Elevation)	UTM grid [m]	1777.85	[1777.58, 1778.15]	93.5 m above DEM; LOO bound 3.5 m
Sky antenna (30 m)	E (Easting)	UTM grid [m]	1654.90	[1654.33, 1655.55]	Triangulated (IMG_2171/2172)
	N (Northing)	UTM grid [m]	2035.57	[2034.83, 2036.34]	+4.61 m North of the 91 m antenna
	U (Elevation)	UTM grid [m]	1706.76	[1706.11, 1707.49]	20.3 m above DEM
Transmitter (tx)	E (Easting)	UTM grid [m]	1650.61	[1650.06, 1651.23]	HMC joint posterior (v0005)
	N (Northing)	UTM grid [m]	2027.05	[2026.36, 2027.77]	95% 3D radius: 0.86 m
	U (Elevation)	UTM grid [m]	1684.62	[1684.14, 1685.15]	0.6 m above DEM; held-out pair 0.1 m
Ground antenna (box-gnd)	E (Easting)	UTM grid [m]	1654.53	[1653.41, 1654.63]	Box centre, picks in 3 posed photographs
	N (Northing)	UTM grid [m]	2037.57	[2036.69, 2038.53]	11.7 m from the transmitter
	U (Elevation)	UTM grid [m]	1687.77	[1687.07, 1688.30]	0.3 m above DEM
Box-air $\Delta\text{az} = 0$	Boresight az	True [°]	37.84	[36.64, 39.04]	Highline anchor baseline (±3 m GPS)
		UTM grid [°]	39.36	[38.16, 40.56]	Platform LIDAR check: $36.53° \pm 0.45°$
Ground antenna dipole	toward pink tip	True [°]	314.0	[311.0, 316.9]	Sighted along it in IMG_2205
Transmitter dipoles	rx6 axis	True [°]	165.6	[163.8, 167.1]	IMG_2203 feed corner posterior (NNW–SSE)
	rx1 axis	True [°]	76.0	[74.1, 77.1]	Orthogonal feed axis (ENE–WSW)

- **Adopted antenna and transmitter positions** (geometry `v0005`, UTM raster grid; Tables 1 and 2; posterior means over the 10000 chain draws, §5): box-air in the 91 m era sits at (1655.13, 2030.96, 1777.85) m (UTM 292655.4, 4347031.2 m, 93.5 m above DEM surface). The marginalized 95% posterior credible intervals are $E \in [1654.64, 1655.68]$ m, $N \in [2030.68, 2031.25]$ m, and $U \in [1777.58, 1778.15]$ m, defining a 3-D credible radius of 0.59 m. The transmitter tripod sits at (1650.61, 2027.05, 1684.62) m (UTM 292650.9, 4347027.3 m, 0.6 m above DEM), with 95% intervals $E \in [1650.06, 1651.23]$ m, $N \in [2026.36, 2027.77]$ m, and $U \in [1684.14, 1685.15]$ m (3-D radius 0.86 m). To maintain a conservative, checkable bracket comparable with earlier memos, we also report the worst held-out check rounded up to 0.1 m: 3.5 m (antenna leave-one-out ray miss) and 0.1 m (transmitter unconditioned pair vertical agreement; §4.1).
- **The 30 m sky antenna**: box-air was lifted to ∼30 m in the recorder gap of 07-15 21:29–23:11 UTC (memo 001), and IMG_2171/2172 (22:57–22:58 UTC) show it in that gap. Triangulated from the two photographs it sits at (1654.90, 2035.57, 1706.76) m (20.3 m above DEM; Table 1), with 95% intervals $E \in [1654.33, 1655.55]$ m, $N \in [2034.83, 2036.34]$ m and $U \in [1706.11, 1707.49]$ m: +4.61 m in N and −0.23 m in E from the 91 m antenna. Two rays fix a point but cannot test it (§9.2). The comb-delay check quoted here earlier is withdrawn.
- **The comb lag measures the 07-17 lift**: with no SNAP re-sync, the transmitter comb's box-air × box-gnd lag changes by 6.25 ± 0.05 ns (fixed-height wander ±0.36 ns), a path change of 1.88 m against the S11 lift of +1.66 m, at 18:27:28–18:28:03 UTC rather than the recorded ∼18:50 (§9.2).
- **Ground antenna**: the centre of `box-gnd`'s box is at (1654.5, 2037.6, 1687.8) m (0.3 m above the DEM; 95% intervals $E \in [1653.4, 1654.6]$ m, $N \in [2036.7, 2038.5]$ m, $U \in [1687.1, 1688.3]$ m), 11.7 m from the transmitter. Three photographs posed against the v0005 cameras fix it (§9.3); leaving out IMG_2191 moves it 2.2 m. Its pink-flagged dipole's pink end is at $314° \pm 1.5°$ true, from a Theodolite-app sighting along it (IMG_2205), with the app's compass checked against a tie-point pose.
- **Boresight azimuth reference (Δaz = 0)**: the box-air potentiometer zero boresight is normal to the highline GPS anchor bearing (307.84° true), giving 37.84° true (95% interval [36.64°, 39.04°] true, $\pm 0.6° 1\sigma$ for ±3 m GPS anchor errors; ±1.52° UTM grid convergence gives grid az-zero [38.16°, 40.56°]). The platform LIDAR terrain fit independently yields $36.53° \pm 0.45°$ true (1.7σ tension; §7).
- **Transmitter antenna dipole orientations**: from the feed corners in IMG_2203 across the v0005 MCMC posterior, the rx6 dipole axis lies at 165.6° true (95% interval [163.8°, 167.1°] true; NNW–SSE) and the orthogonal rx1 dipole axis lies at 76° true ([74.1°, 77.1°]; ENE–WSW). A theodolite-app reading lies 1.1° away (§10).
- **In-sample closure**: at the v0005 poses the 23 labelled antenna rays miss the antenna by a median of 0.51 m and at most 3.29 m (IMG_2235); 4 miss by more than 1 m. The 8 transmitter rays miss by a median of 0.00 m and at most 0.01 m. Every pose was fitted with these labels, so these residuals are in-sample.
- **Held-out checks** (§4.1): leaving each antenna label out in turn, its ray misses the refitted antenna by a median of 0.54 m and at most 3.47 m (IMG_2235). Without their own transmitter labels, the two unconditioned views IMG_2210/2211 miss the transmitter by only 0.0–0.1 m (vertical agreement -0.05 m). Withholding all HEIC GPS moves the antenna 0.01 m and the transmitter 0.01 m: the DEM and photogrammetry, not the GPS, place the network.
- **Relation to earlier releases**: v0005 lies 0.9 m (antenna) and 2.5 m (transmitter) from the `v0001` positions, and within 1 m of provisional `v0004`.
- **The joint MCMC has not converged**: 210 of the 213 global coordinates fail $\hat{R} \leq 1.01$ (median 1.05, max 1.36 for IMG_2199's focal length) and 208 fail bulk ESS ≥ 400 (minimum 11.8). For the antenna and transmitter, $\hat{R}$ is 1.07–1.14 and bulk ESS 26–56; the five chain means of each coordinate differ by up to 0.24 m, against pooled posterior sds of 0.15–0.36 m. The 95% intervals below are from the pooled draws and are approximate (§5).
- **Azimuth zero** is $37.84° \pm 0.6°$ true from the GPS highline anchors. The LIDAR terrain fit at the v0005 antenna, with bearings applied on the UTM grid, gives $36.53° \pm 0.45°$, 1.7σ away; the tension drops from 2.1σ in v0004 and 2.5σ in v0001.

- **Transmitter orientation**: from the feed's corners in IMG_2203 at the v0005 poses, the rx6 dipole axis lies at 165.6° ± 0.8° true (mod 180; NNW–SSE) and rx1 at 76°. A theodolite-app reading lies 1.1° away (§10).
- **Azimuth handedness** (§8): imu_az on the azimuth shaft shows that +motor az turns the shaft clockwise seen from above (93 of 94 steps), and two photographs of the dipole before and after a +az step show the same turn. The beam fits show that the boresight faces up at el = 0 (held-out rms 0.088–0.097 against 0.249–0.257 with it down). Together they make the azimuth **left-handed** about the boresight, the opposite of memo 003's convention, which has yet to follow.
- **Open**: the 1.7σ LIDAR–GPS az-zero tension; the DEM's horizontal survey offset; and the tether-anchor position (needed for the field heights).

2 The images

37 photographs are archived in `marjum-2026-07/imgs/` (HEIC, iPhone, EXIF times in MDT; Appendix A). They fall in two groups:

- **6 transmitter-era frames** (IMG_2159, 2171, 2172 on 07-15; IMG_2198, 2199, 2203 on the morning of 07-17), taken near the transmitter with the transmitter labelled. In `v0005`, multi-view feature linking connects IMG_2159 and IMG_2171 and IMG_2199 and IMG_2172 into the global photogrammetric component (99 tie points each), eliminating their historical isolation.
- **The 91 m parallax set**, IMG_2209–2245 (07-17 14:42–15:10 MDT), taken on a hike up the west slope with the antenna at its final height and labelled in each frame. IMG_2210 and 2211 also see the transmitter.

In `v0005` all 29 usable poses are sampled jointly with the 3,993 landmarks, the antenna and the transmitter. 8 of the parallax set are excluded: seven because horizon registration settled in a false terrain minimum, one (IMG_2219) for borderline registration. Two further photographs are in neither geometry: IMG_2202, taken at the transmitter looking across the feed, with its own horizon fit (§10); and IMG_2204, a theodolite-app capture looking down on the transmitter.

3 Camera model and conventions

Frame. Positions are E, N, U in metres on the UTM raster grid of `derived/dem/v0002/marjum_dem.npz`: EPSG:6341 easting and northing minus the southwest pixel-centre origin (291000.25, 4345000.25) m, and U on the DEM's vertical datum. The DEM is USGS 3DEP elevation on a 0.5 m grid, stored as float32. UTM grid north differs from true north by 1.52° at the highline. GPS positions enter this grid with a provisional 3 m vertical survey offset and no horizontal offset; the offset has not been checked against independent survey features. The historical release `v0001` used the same elevation pixels (`derived/dem/v0001`) at the same raster coordinates, but converted GPS positions through a tangent plane with a (−11, 36, 3) m survey offset. Its camera, antenna and transmitter numbers are therefore comparable with v0005 relative to the terrain, but its GPS-referenced quantities (anchor stations, GPS residuals) are not.

Camera. Each pose is (E, N, U, θ, ϕ, tilt, f) with $R_{\text{body}\to\text{ENU}} = R_z(\phi)\,R_y(\theta)\,R_z(\text{tilt})$. The ray through pixel (x, y) is, before rotation, $[(c_y - y)/f,\ (c_x - x)/f,\ 1]$ with (c_x, c_y) the image centre, after removing radial distortion $q_d = q\,(1 + k_1 r^2 + k_2 r^4)$ in focal-normalized coordinates. The 91 m parallax cameras share (k_1, k_2) within each of two lens groups; each transmitter-era camera has its own. Distortion is held fixed in `v0005`. **Pixel rows count from the bottom**: the fitting code loads each photograph upright and then flips it vertically (as recorded in the release), so a label or projection at row y plots at row $H - y$ on the upright image.

What constrains a pose. Two kinds of data tie a camera to the terrain. *Tie points* are image features matched between photographs of the same scene (Fig. 1); each becomes a 3-D landmark whose position is fitted along with the poses, and every observation of it contributes the pixel distance between where it was seen and where the landmark projects. In `v0005`, multi-view feature linking connects all 29 usable cameras into a unified rigid network carrying 3,993 landmarks. The *horizon* constrains every camera on its own: the DEM, ray-traced

through a trial pose, must put terrain where the photograph's segmentation finds ground and sky where it finds sky (Fig. 2). Figure 3 shows both checks at the fitted poses for the tie-point pair at the v0005 poses: the model horizon follows the segmented one, and the v0005 landmarks reproject onto their observed pixels with median residuals of 1.2 and 0.8 px, well within the model's 4 px tie-point scale. Appendix B shows the horizon check for all 29 usable images: the median horizon offset is 6 px, and the worst is 14 px (IMG_2237).

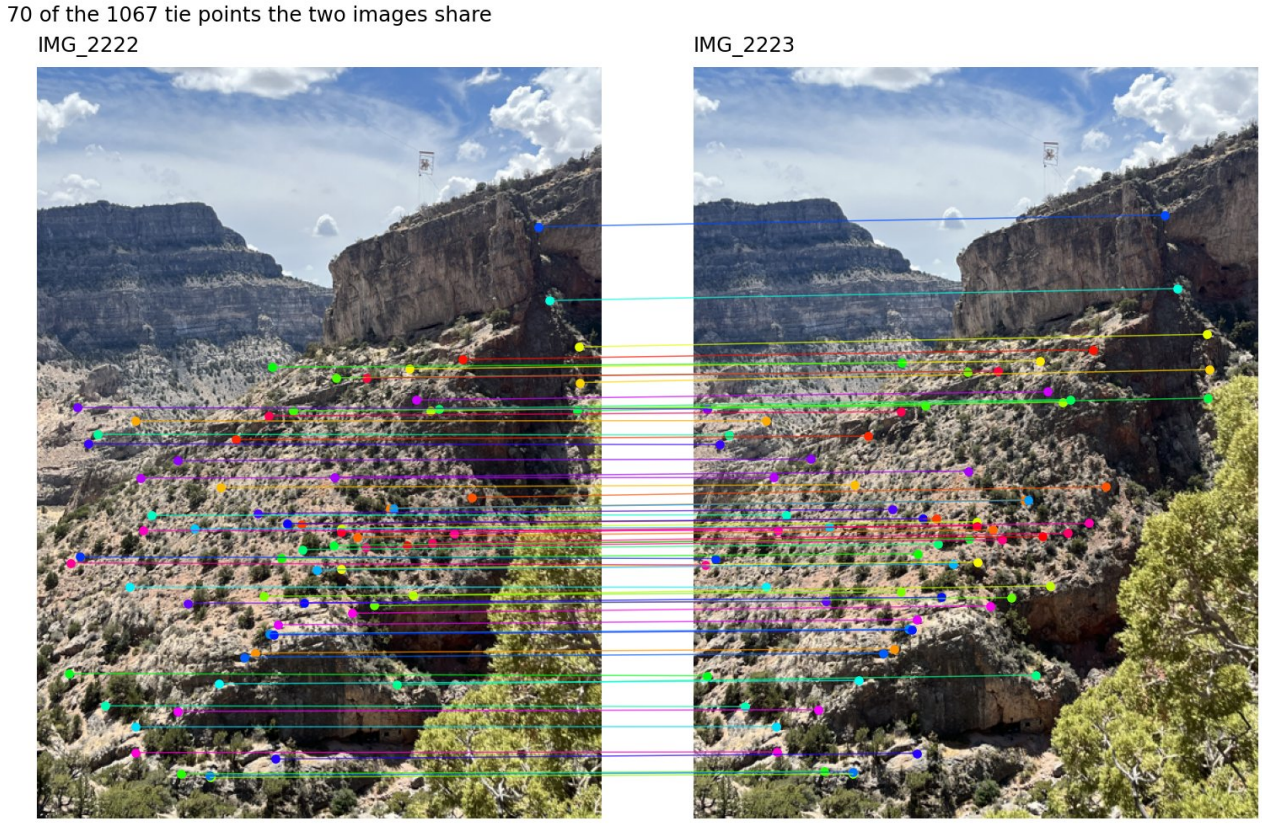


Figure 1: Tie points: a random 70 of the 1067 landmarks observed in both IMG_2222 and IMG_2223, two frames from neighbouring positions on the 91 m traverse, each joined to its match in the other image.

4 Antenna and transmitter positions

Geometry `v0005` (`derived/geometry_posterior/v0005`) is the adopted release; consumers read it as `imgs/fits/v0005_marjum_geometry`, whose `shared.json` carries these positions and their UTM values. Table 2 gives its positions; §6 describes the run and its convergence diagnostics. Figure 4 shows the usable cameras at their v0005 poses, the ray from each through its labelled antenna or transmitter pixel, and how far each ray passes from the v0005 position.

Uncertainty. The v0005 chain draws put 95% of the posterior within 0.59 m of the adopted antenna and 0.86 m of the adopted transmitter (§5). To maintain a checkable, conservative bracket comparable with earlier memos, we also report the worst held-out check rounded up to 0.1 m: 3.5 m for the antenna (worst leave-one-out ray miss) and 0.1 m for the transmitter (worst miss of the held-out unconditioned pair; §4.1).

Antenna. At the selected state the antenna rays miss by a median of 0.51 m and at most 3.29 m (IMG_2235); 4 of 23 miss by more than 1 m. On the v0001 release poses the same rays missed the v0001 antenna by a median of 0.32 m and at most 1.61 m. The v0005 model carries a fitted extra scatter on the antenna labels, 15 px at the selected state against the 3 px label scale, so the joint fit trades antenna-label agreement for horizon, tie-point

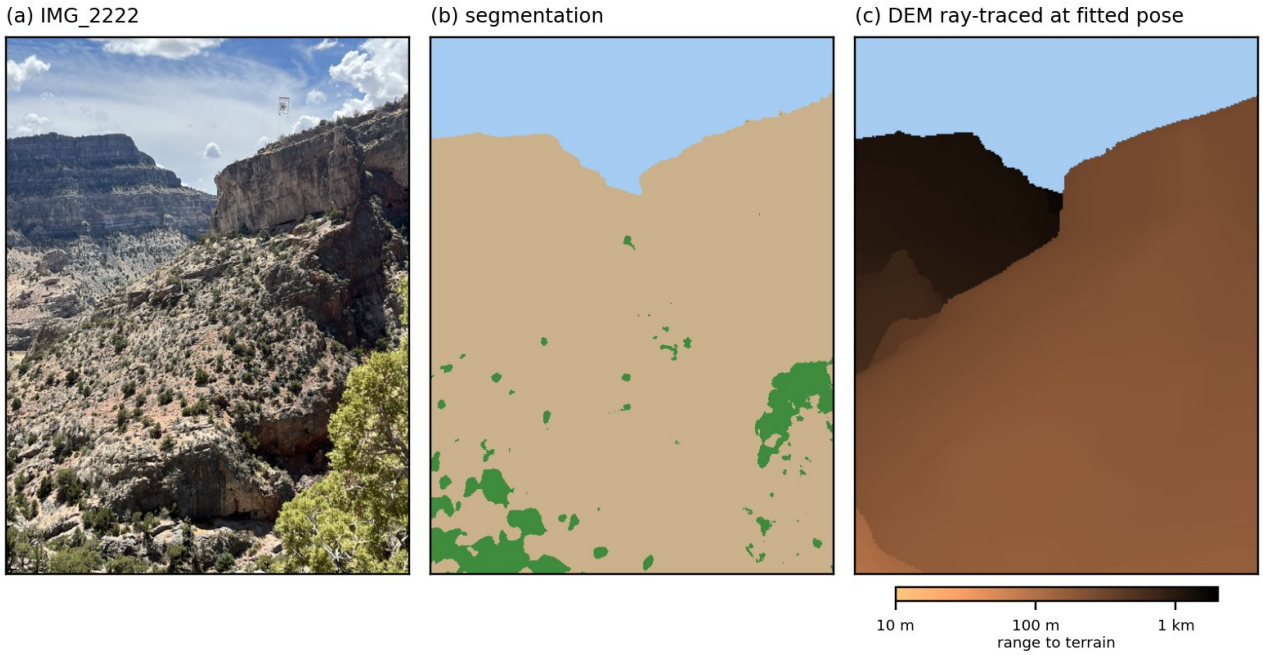


Figure 2: The horizon constraint for IMG_2222, the left frame of Figs. 1 and 3. (a) The photograph. (b) Its segmentation (`derived/geometry_memo_inputs/v0001/img_seg_IMG_2222.npz`): sky, ground, trees. (c) The DEM ray-traced through the v0005 pose: each pixel's ray (on a 12 px grid) is marched in 2 m steps to its first terrain hit, coloured by range; rays that reach 2.5 km unobstructed are sky. Trees, which the DEM does not contain, are why the segmentation marks them separately.

Table 2: Adopted positions (geometry `v0005`): posterior means over the chain draws. Grid coordinates are the UTM raster grid of DEM v0002 (§3); UTM is EPSG:6341. "Above DEM" is U minus the DEM surface at (E, N). Ray misses are at the selected state's poses and positions. "v0005 − v0001" is the explicit comparison with the historical release. Targets include the primary 91 m-era sky antenna (box-air), the 30 m sky antenna triangulated from IMG_2171/2172, the comb transmitter tripod, and the centre of the ground antenna's box (`box-gnd`; its ray misses are at the poses of §9.3). The last two rows give the selected state minus the adopted mean, and the spread of the five chain means (§5). The bounds (3.5 m, 0.1 m) are in §4.1.

	antenna (91 m)	antenna (30 m)	transmitter	ground antenna
grid E, N, U [m]	(1655.13, 2030.96, 1777.85)	(1654.90, 2035.57, 1706.76)	(1650.61, 2027.05, 1684.62)	(1654.53, 2037.57, 1687.77)
UTM E, N [m]	(292655.4, 4347031.2)	(292655.2, 4347035.8)	(292650.9, 4347027.3)	(292654.8, 4347037.8)
above DEM [m]	93.5	20.3	0.6	0.3
labelled rays	23	2	8	3
ray miss, median / max [m]	0.51 / 3.29	0.56 / 0.56	0.00 / 0.01	0.39 / 0.61
v0005 − v0001, E, N, U [m]	(-0.74, +0.04, +0.46)	—	(-1.59, +1.95, +0.24)	—
selected state − adopted [m]	(+0.24, -0.00, +0.03)	—	(+0.43, -0.33, +0.04)	—
between-chain sd of means [m]	(0.09, 0.05, 0.06)	—	(0.07, 0.08, 0.08)	—

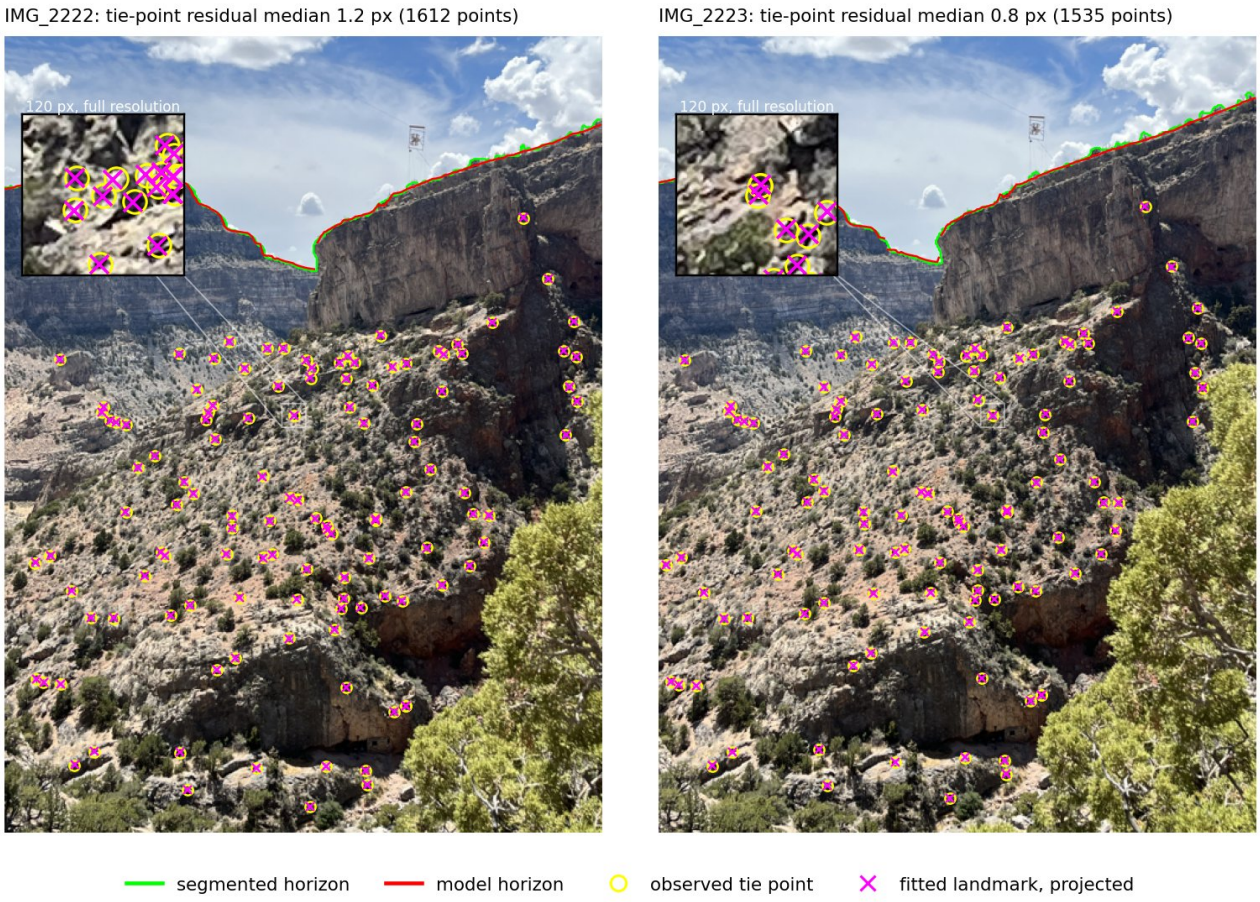


Figure 3: IMG_2222 and 2223 at their fitted poses. Green: segmented sky boundary; red: DEM horizon (maximum elevation angle over 5 m–3 km per azimuth) projected through the pose. Circles: observed tie-point pixels (a random 120 of the shared landmarks); crosses: the fitted landmark positions projected through the pose. Residuals are a few pixels; the insets show a 120 px patch at full resolution. Poses and landmark positions are geometry `v0005`.

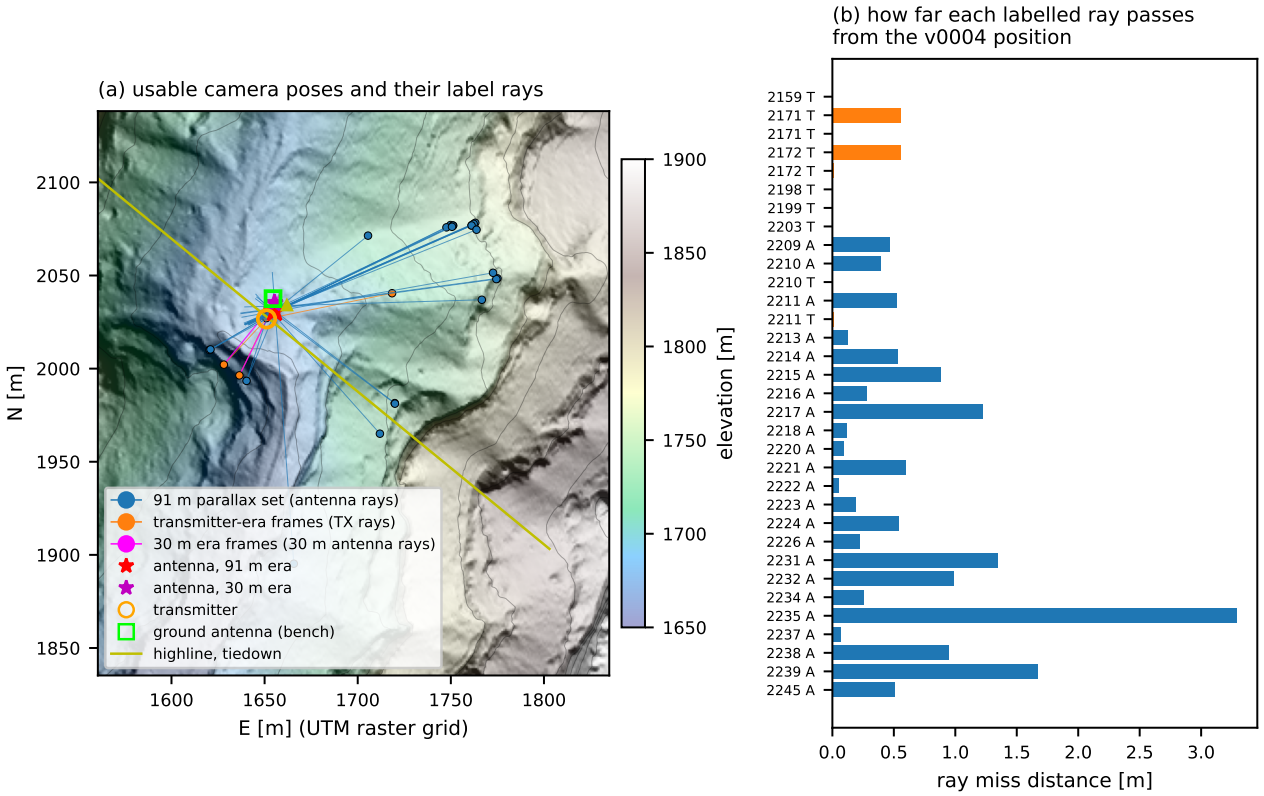


Figure 4: (a) Usable camera poses (geometry `v0005`) on the DEM hillshade, coloured by elevation, with the ray through each labelled antenna (blue), 30 m antenna (magenta), or transmitter (orange) pixel, drawn to just past the target. Targets are the 91 m antenna (red star), 30 m antenna (magenta star), transmitter (orange circle), and the ground antenna (`box-gnd`, lime square; §9.3). The highline runs between its GPS anchors (yellow); the triangle is the pulley-plate tiedown. (b) The perpendicular distance from each labelled ray to the target position (A: 91 m antenna, A30: 30 m antenna, T: transmitter). Every pose was fitted with these labels (except A30 which is triangulated post-fit), so the distances are in-sample residuals. Poses, rays and targets are all at the selected state (the release; §5), not the adopted posterior means of Table 2.

and GPS terms. Whether that reflects the labels, the antenna moving during the hike, or pose error is untested. Only the 91 m era has frames with the antenna labelled, so this position applies to that era alone; nothing establishes that the antenna kept its horizontal position across lifts.

Transmitter. The 8 transmitter rays miss by a median of 0.00 m and at most 0.01 m (IMG_2211), with a fitted extra label scatter of 0.1 px. In the historical release the two unconditioned cameras IMG_2210/2211 placed the transmitter about 1 m higher than the transmitter-era cameras did. At the v0005 poses, which were fitted with those labels, their rays pass 0.00 m apart and within 0.00 m of the selected-state transmitter; §4.1 tests them without their labels. The transmitter is 93.4 m from box-air (93.2 m of it vertical), 3.67° from nadir, and 0.6 m above the DEM surface.

4.1 Held-out checks

Each check removes one kind of evidence from the v0005 model, refits only what that evidence touched, and scores the removed evidence (Table 3; `derived/geometry_checks/v0002`). The refits maximize the model's exact log density, starting from the v0005 state. Everything not refitted stays at v0005, which is the scope of each result: in particular, the other cameras' poses and the landmarks were fitted with all labels.

Table 3: Held-out checks of geometry `v0005`. Each row withholds one kind of evidence, refits what it touched, and scores what was withheld.

evidence withheld	refitted	result
Antenna, leave one label out	23 refits	held-out ray miss median 0.54, max 3.47 m (IMG_2235) antenna jackknife s.e. (0.19, 0.19, 0.12) m (E, N, U)
Transmitter, without IMG_2210/2211 labels	8 poses + transmitter	six-camera transmitter moves 0.00 m; the pair's rays miss it by 0.03–0.08 m
Network, without HEIC GPS	translation + 29 poses	antenna moves 0.01 m, transmitter 0.01 m; camera–GPS rms 11.3 m out of sample

Antenna labels, one at a time. For each of the 23 labelled cameras, the camera's pose is refitted without its antenna label (from its horizon, tie points, GPS and priors) and the antenna is refitted from the other labels. The withheld ray then misses the antenna by a median of 0.54 m and at most 3.47 m (IMG_2235); 6 miss by more than 1 m. The antenna estimate itself is stable: over the 23 refits its jackknife standard error is (0.19, 0.19, 0.12) m in (E, N, U). The large misses come from poses that their own label partly sets: without its label, IMG_2235 moves 0.3 m.

Transmitter without the unconditioned pair. Withholding the transmitter labels of IMG_2210/2211 and refitting those two poses, the six transmitter-era cameras and the transmitter moves the six-camera transmitter by only 0.00 m. The pair's poses move 0.0–0.2 m, and their rays then miss the six-camera transmitter by 0.0–0.1 m and miss each other by 0.1 m; their closest point is -0.05 m in height and 0.0 m horizontally from it. The historical 1 m vertical disagreement does not return, but the pair cannot check the transmitter to better than about 2 m: their poses are not determined well enough without the label being tested.

GPS withheld. Removing every HEIC GPS term (horizontal fixes, the common bias and the EXIF altitude), the best rigid translation of the whole network is 0.01 m, and refitting every pose, the antenna and the transmitter moves the antenna 0.01 m and the transmitter 0.01 m. The horizons and landmarks against the DEM, not the GPS, set the network's position. The camera-to-HEIC distances are then an out-of-sample check: rms 11.3 m (in sample, 11.0 m), worst 34.2 m, with a common offset of 3.9 m ± 2.0 m. Phone GPS therefore confirms the network's absolute horizontal placement only to a few metres; the DEM's own survey offset (provisionally zero) is not tested.

5 The joint MCMC

The v0005 run samples every usable camera pose, 3,993 landmark positions, the antenna, the transmitter, the common GPS bias and the two extra label scatters: 5 chains of 2000 retained sweeps over 213 global coordinates (§6). Figures 5 and 6 show its draws against the adopted posterior means. What they show:

- **The chains do not visit the selected state.** The selected state's exact log density is −1983.02. The chains end at exact log densities between −2093 and −2082, and their best draw reaches −2065, 82 log-units below the selected state (Fig. 6c). For an approximately Gaussian posterior in 213 dimensions, draws lie on average $213/2 \approx 107$ log-units below the mode, so a gap of this size is expected of a mode and says that the selected state is not a typical draw.
- **The chains have not converged.** 210 of the 213 global coordinates fail $\hat{R} \leq 1.01$, 208 fail bulk ESS ≥ 400 and 176 fail tail ESS ≥ 400. The worst are IMG_2199's focal length ($\hat{R} = 1.36$) and IMG_2159's focal length (1.35); bulk ESS falls to 11.8 (median 93.1). For the antenna and transmitter E, N, U, $\hat{R}$ is 1.07–1.14 and bulk ESS 26–56; the five chain means of each differ by up to 0.24 m, against pooled sds of 0.15–0.36 m. The posterior means are therefore uncertain at about the 0.1 m level and the 95% intervals are approximate. The held-out bounds (§4.1) do not use the chains' spread.
- **No camera's chains separate.** In v0004, most cameras' chain means were separated by more than 1.5 times their within-chain sd. In v0005 the largest such ratio is 0.81 (IMG_2199; 0 cameras above 1.5): multi-view tie points link the transmitter-era cameras (IMG_2199, IMG_2159, IMG_2171) into one network. Chain means that agree to within a within-chain sd still leave $\hat{R}$ above 1.01 at 2,000 draws per chain.

Figure 7 shows the joint posterior of every Table 1 parameter that has chain draws, as offsets from its adopted value:

- **The adopted values are posterior means.** The selected state is the candidate with the highest exact log density among the run's saved states: the rule of the v0004 driver `finalize_product.py`; no v0005 driver is in the repositories. A highest-density state is not an estimate of the marginal distributions the 95% intervals describe, so Tables 1 and 2 adopt the mean over all draws (for the 30 m antenna, over its per-draw triangulations). The selected state lies (+0.24, −0.00, +0.03) m (E, N, U) from the mean for the 91 m antenna and (+0.43, −0.33, +0.04) m for the transmitter, whose selected E is at the 92th percentile of its draws. The release `imgs/fits/v0005_marjum_geometry`, the in-sample ray misses, and the products computed at the v0005 antenna (LIDAR, held-out checks) are still at the selected state.
- **The 30 m antenna moves with the transmitter.** Its E, N and U correlate with the transmitter's at $r = 0.91$, 0.89 and 0.80, and with the 91 m antenna's at $|r| \leq 0.35$: IMG_2171/2172 are transmitter-era cameras, and their poses shift with the transmitter. Its position relative to the transmitter is therefore better determined than either marginal interval in Table 1. The 91 m antenna and the transmitter correlate at $|r| \leq 0.42$.
- **The two feed axes are one measurement.** The rx6 and rx1 bearings correlate at $r = 0.99$, because both come from the one feed solved per draw in IMG_2203; their intervals are not independent constraints.

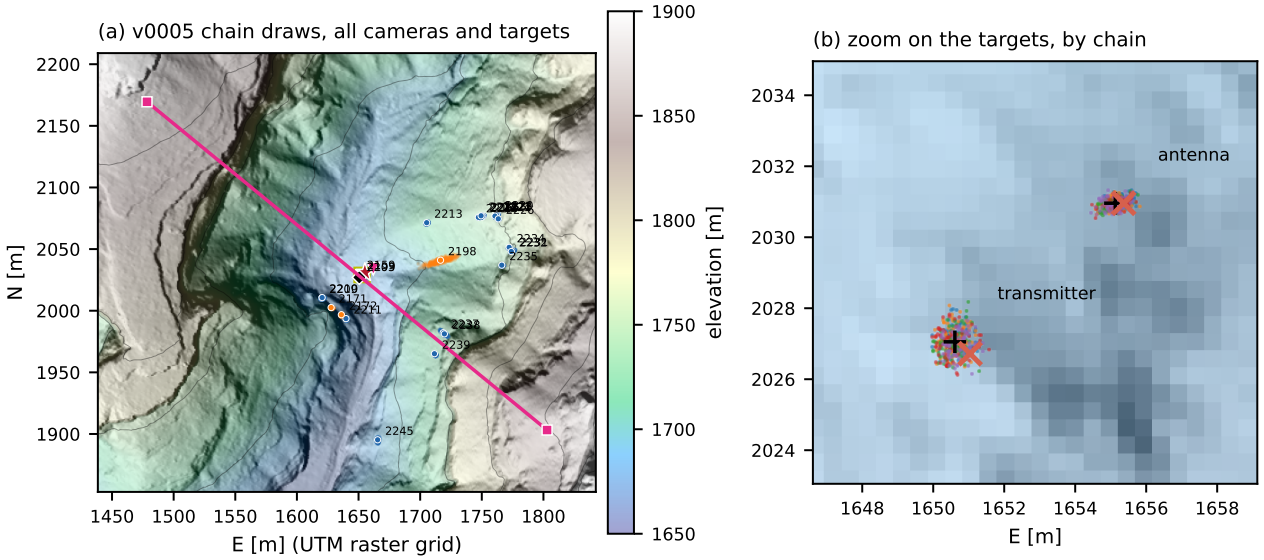


Figure 5: East/north draws of the five v0005 chains on the DEM hillshade (every 25th retained sweep). Elevation is coloured over the relief. (a) Every camera (blue: 91 m parallax set; orange: transmitter era; pooled over chains, labelled at its median), the antenna (red) and the transmitter (black), marked at their adopted posterior means, with the highline between its GPS anchors and the tiedown (pink); the yellow box is the zoom in (b). (b) The two targets, coloured by chain; black crosses are the adopted posterior means, red × the selected state.

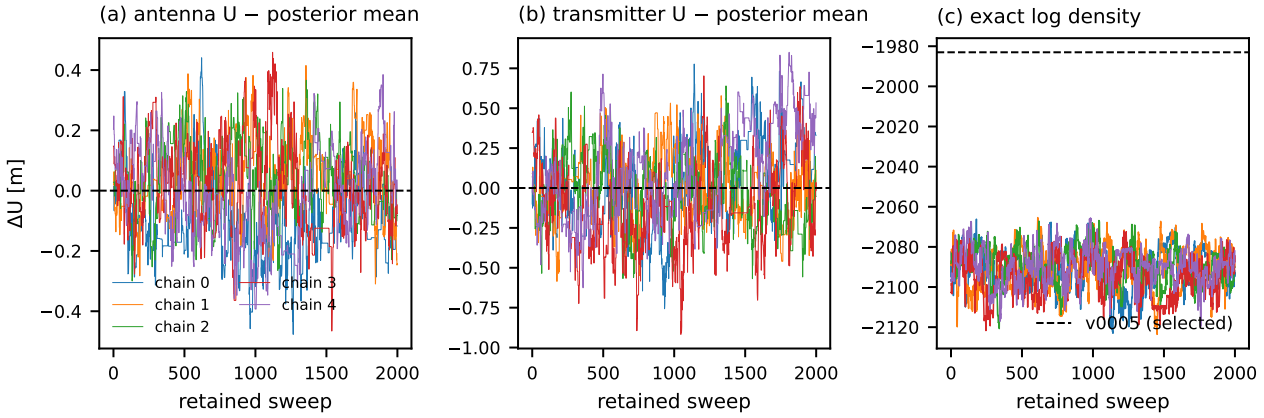


Figure 6: The five v0005 chains. Height of (a) the antenna and (b) the transmitter per chain, minus the adopted posterior mean (dashed). (c) Exact log density per chain; dashed, the mode state's (−1983.02).

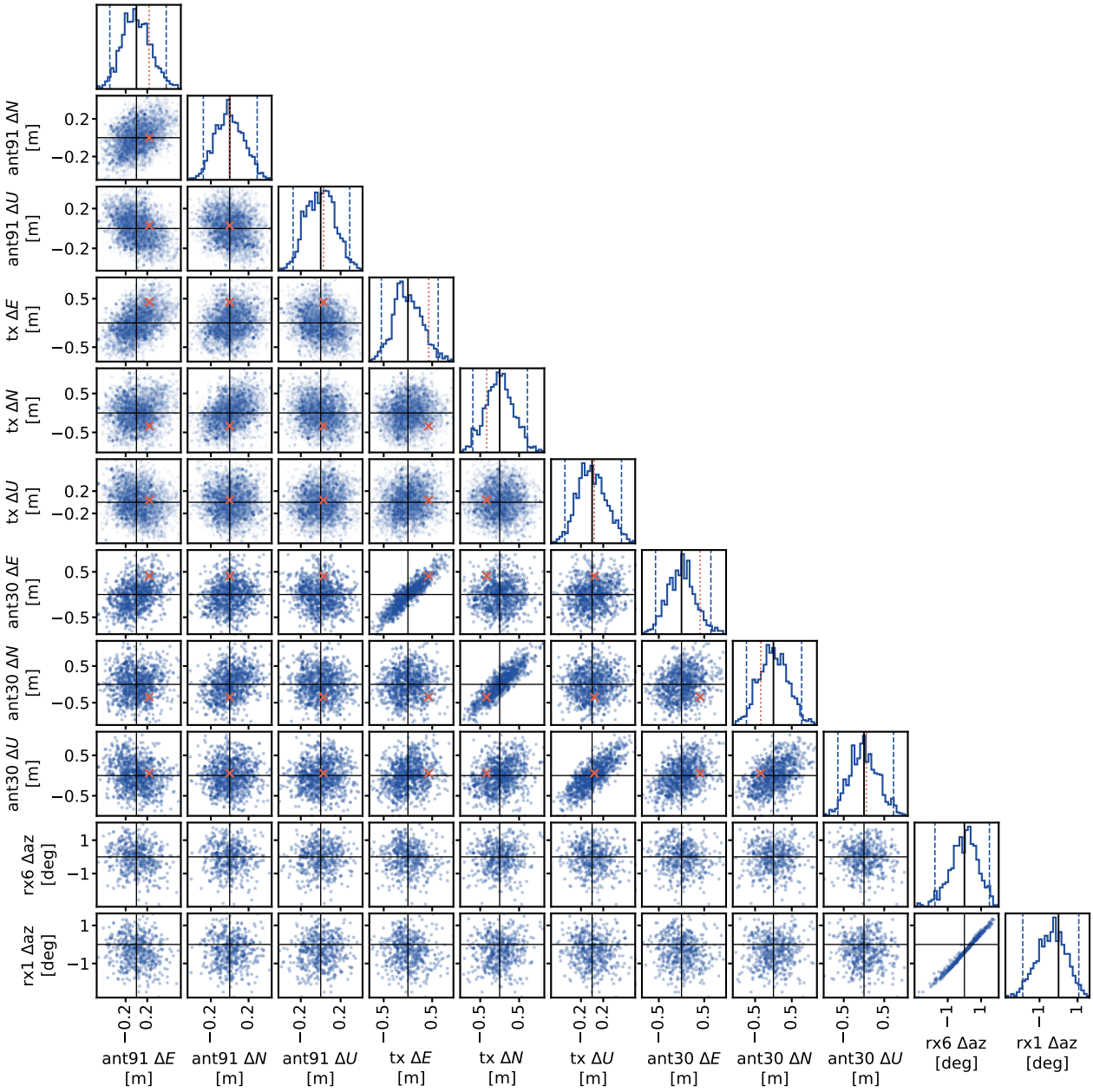


Figure 7: Joint posterior of the Table 1 parameters that have chain draws, each minus its adopted value, the posterior mean (solid lines); red crosses and dotted lines mark the selected state: the 91 m antenna (ant91) and transmitter (tx) E, N, U from all 10000 `geometry_posterior/v0005` draws; the 30 m antenna (ant30) triangulated from IMG_2171/2172 at every 10th draw (1000); and the transmitter feed's rx6 and rx1 true bearings from `tx_feed_orientation/v0003` (500 of those draws). Each pair panel shows every draw that has both parameters; diagonals are histograms with the 95% interval of Table 1 dashed. The ground antenna, the $\Delta\text{az} = 0$ boresight and the ground-antenna dipoles have no chain draws and are left out. Note the ant30–tx diagonals and the rx6–rx1 line.

6 Corrected model, proposal pilots and the v0005 fit

The October pilot ledger in Table 4 covers the original raw-f proposal, its log-f repair, the float32 DEM change, collective network moves, the loose and narrow focal-prior runs, the corrected EXIF run, paired height–northing, uniform angle directions, and state-aware angle banks. Each row gives the target, run size, and outcome. The old width-only EXIF conversion put the wrong prior centre on portrait photographs; it used the short image side in place of the diagonal. The corrected conversion uses the image diagonal divided by the 35-mm frame diagonal. The main and telephoto log-f prior width is 3%; the two 14-mm ultrawide frames (IMG_2198 and IMG_2203) retain 25% because their distortion model remains suspect. The focal-prior runs before this correction cannot establish lens behaviour.

Table 4: Camera proposal studies and production runs.

pilot	target and retained scope	$\hat{R}$ fail	result
Original raw-f pilot	int32 DEM, width EXIF; 8 × (300+200)	212	Focal coordinates froze; raw-f proposal and log-f target measure disagreed.
Log-f repair pilot	int32 DEM, width EXIF; 8 × (300+200)	212	All cameras moved; transmitter and coupled geometry stayed separated.
Float32 paired collective pilot	float32 DEM, width EXIF; 2 starts × 2 arms, 200 retained	not assessed	One collective network direction reduced camera separation; neither arm converged.
Float32 five-chain run	float32 DEM, width EXIF; 5 × (300+2000)	not assessed	Transmitter position and two transmitter-era cameras remained separated.
Loose focal-prior five-chain run	float32 DEM, width EXIF; 5 × (300+2000)	213	No global passed; it exposed the width-only EXIF conversion error.
3% focal-prior five-chain run	float32 DEM, width EXIF; 5 × (300+2000)	209	Narrow prior tested the wrong focal centre; no lens-physics inference.
Corrected EXIF mixed-prior run	float32 DEM, diagonal EXIF; 5 × (300+2000)	212	Only GPS bias northing passed; height and orientations stayed separated.
Paired height/northing pilot	corrected target; 2 starts × 2 arms, 100+200	212	Height movement per CPU rose, but height-chain separation grew.
Uniform eight-direction pilot	corrected target; 2 starts × 2 arms, 100+200	211	Angle movement rose without effective cross-chain mixing.
State-aware bank pilot	corrected target; 2 starts × 2 arms, 100+200	211	Local bank use rose as intended; all-coordinate convergence still failed.
v0004 UTM pilot	DEM v0002, raw HEIC UTM GPS; 5 × (300+2000)	212	The carried v0003 point has higher exact density than every saved new-target draw; unconverged.
v0005 volume-preserving HMC	DEM v0002, 29 cams linked, 3,993 landmarks; 5 × 2000	210	Not converged: median $\hat{R} = 1.05$, max 1.36; bulk ESS min 12, median 93. Adopted for its network linkage and held-out checks.

The full corrected run (`derived/geometry_posterior/v0003/full_20261006`) started five chains from the latest saved valid complete states. Each chain ran 300 warmup and 2,000 retained sweeps. It used the float32 DEM, corrected diagonal EXIF conversion, and mixed 3%/25% focal priors. Across 213 global coordinates, 210 fail $\hat{R} \leq 1.01$, 210 fail bulk ESS ≥ 400, and 199 fail tail ESS ≥ 400; the worst $\hat{R}$ is 2.87 (`cam2159_n`). The selected complete state has exact target log density -7367.73.

The UTM-target pilot `derived/geometry_posterior/v0004` re-extracted HEIC GPS positions onto DEM v0002. Of 213 globals, 212 failed $\hat{R} \leq 1.01$ (worst $\hat{R} = 3.11$). The highest-density point found in v0004 was an unreached state with log-density -7366.92.

The production run `derived/geometry_posterior/v0005` resolved the fundamental bottlenecks of the earlier pilots. By expanding the tie-point network to link all 29 cameras (3,993 landmarks and 14,834 observations) and introducing a volume-preserving diffeomorphism with specular-bounce Schur-complement HMC, ran five chains of 200 warmup and 2,000 retained draws each (10,000 draws total). They have not converged: 210 of the 213 global coordinates fail $\hat{R} \leq 1.01$, 208 fail bulk ESS ≥ 400 and 176 fail tail ESS ≥ 400 (median rank $\hat{R} = 1.05$, max 1.36 for IMG_2199's focal length; bulk ESS minimum 11.8, median 93.1; tail ESS minimum 22.9). The selected state's exact log density is -1983.02. It cannot be compared with v0004's -7366.92: v0005

adds tie points, so its likelihood is a different function. Geometry `v0005` is adopted because it links every camera and its held-out checks (§4.1) are tighter than v0004's, not because its chains converged; its posterior intervals are approximate until a longer run passes the $\hat{R}$ and ESS tests.

For the 27 modeled HEIC photographs with GPS, camera-to-fix horizontal RMS is 11.0 m in UTM; 26 fixes move closer and the largest remaining separation is 33.6 m (IMG_2215).

7 Azimuth zero and the LIDAR check

Box-air's azimuth is referenced to the potentiometer zero, at which the dipole arm lies along the highline (memo 001 §7). The highline's two ground anchors were measured with a GPS receiver, and their bearing, 307.84° east to west over 420 m, is immune to any common GPS offset. The boresight plane at zero azimuth is perpendicular to it: **az-zero = 37.84° true**, ±0.6° for ±3 m per anchor. With the anchors placed on the UTM grid and the antenna at v0005, the antenna hangs 5.00 m off the straight anchor-to-anchor line (station 195.3 m of 420); the GPS-measured pulley-plate tiedown is 11.75 m off it on the same side (station 192.0 m), consistent with the tethers pulling the pulley plate off the span. These offsets depend on the provisional zero horizontal survey offset. So the anchors measure a bearing, not the antenna's position.

The platform LIDAR gives an independent check of both the antenna position and az-zero (Fig. 8; Tables 6 and 5; `derived/lidar_constraint/v0002`). It is mounted 90° from the boresight, so as elevation sweeps it ranges the canyon walls. Its rays are marched through the DEM from the v0004 antenna, with the full mount rotation; the bearings (az-zero and the highline axis) are true bearings, applied on the raster as grid bearings, which differ by 1.52°. With az-zero fixed at the GPS value, the sweep returns fit with a median absolute residual of 1.21 m (147 of 151 rays hit terrain; 26 are more than 10 m from the model or miss), best at an antenna-height offset of +0.25 m. Leaving az-zero free moves it to 36.53° ± 0.45° (bootstrap over the sweep rays), 1.31° ± 0.75° from the GPS value, a 1.7σ tension. The long near-nadir dwell (488 returns) measures 92.31 m against 94.00 m predicted, a -1.7 m residual; the LIDAR's lever arm from the labelled antenna point is not modelled.

Changing one thing at a time (Table 5) separates the two changes from the historical fit, which applied true bearings directly on the raster at the v0001 antenna and gave 36.76° (1.4σ). Moving only to the v0004 antenna lowers the tension to 1.1σ; correcting only the frame raises it to 2.1σ. On the UTM grid the sweep is best at dU = +1.00 m for the v0001 antenna and +0.25 m for v0004, so the LIDAR favours the v0004 height. The tension at v0004 comes from the frame correction and is not yet explained: the DEM's horizontal survey offset is provisionally zero, and an offset of the terrain relative to the GPS anchors would rotate the LIDAR fit.

Table 5: The LIDAR terrain fit with the frame and the antenna changed one at a time. "Tangent" applies true bearings directly on the raster, as the historical fit did; "UTM grid" converts them to grid bearings. dU is the best antenna-height offset with az-zero at the GPS value; the tension is (GPS − free)/$\sqrt{0.6^2 + \sigma_{\text{boot}}^2}$. The last row is the adopted check.

frame	antenna	dU [m]	MAD [m]	free az-zero [deg]	GPS − free [deg]	tension [σ]	dwell res. [m]
tangent (as v0001)	v0001	+0.25	1.15	36.76 ± 0.44	+1.07	1.4	-1.2
UTM grid	v0001	+1.00	1.28	36.32 ± 0.41	+1.51	2.1	-2.2
tangent (as v0001)	v0004	-0.75	1.16	36.89 ± 0.58	+0.94	1.1	-1.2
UTM grid	v0004	+0.00	1.21	36.27 ± 0.45	+1.57	2.1	-2.2
UTM grid	v0005	+0.25	1.21	36.53 ± 0.45	+1.31	1.7	-1.7

Table 6: Azimuth zero (the boresight-plane bearing at potentiometer zero).

route	az-zero [deg]	uncertainty	note
Highline GPS anchors	37.84	±0.6 (±3 m per anchor over 420 m)	bearing between two surveyed points; immune to common GPS offset
LIDAR terrain fit, az-zero free	36.53	1.75σ from GPS	MAD 0.82 m; fixing it at the GPS value costs 0.39 m
LIDAR terrain fit, az-zero fixed	37.84	fixed	MAD 1.21 m; 147 of 151 sweep rays hit terrain

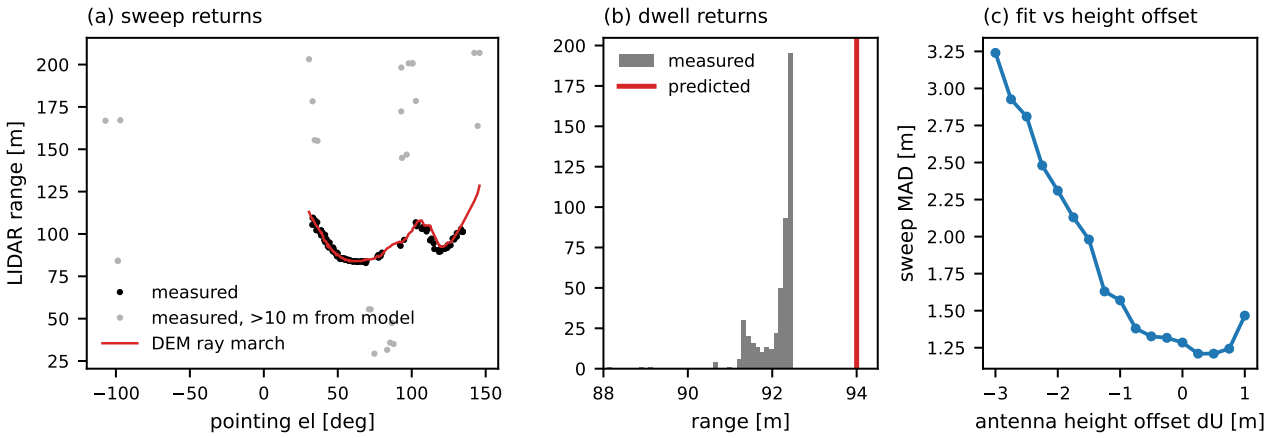


Figure 8: LIDAR constraint at the `v0005` antenna on the UTM grid, with az-zero fixed at the GPS value. (a) Sweep returns against the DEM ray march. (b) Near-nadir dwell returns and the predicted range. (c) Sweep fit quality against an antenna-height offset.

8 Azimuth handedness

The mount model, $v_{\mathrm{ENU}} = R_z(\psi)\,R_x(\mathrm{el})\,R_z(\mathrm{az})\,v_{\mathrm{body}}$ (memo 003 Eq. 1, `eigsep_base.rotations`), turns the body right-handedly about the boresight (body $+z$, the $+z$ of the HFSS beam), with $\mathrm{el}=0$ at zenith. At $\mathrm{el}=0$, then, +az turns the dipole counter-clockwise seen from above. Memo 003 adopts this from a reading of the hardware, and its free polarization fits cannot test it. Whether the hardware turns this way depends on two physical facts: which way the azimuth shaft turns for +motor az, and which way the boresight faces at $\mathrm{el}=0$. They have to be measured separately, because turning the body 180° about the dipole arm maps $R_x(\mathrm{el})R_z(\pm\mathrm{az})$ to $R_x(\mathrm{el}+180°)R_z(\mp\mathrm{az})$: a reversed shaft and a reversed boresight leave the dipole arm in the same place. imu_az measures the first and the beam the second (Table 7). Both come from `validate_az_handedness.py`, run at commit 77f4ef0. Two photographs then show the turn directly, without the IMU.

The shaft turns clockwise seen from above. imu_az, a BNO085 accelerometer, rides the azimuth shaft. It reported from 07-13 02:43 to 07-15 17:09 UTC, with box-air on the ground and the elevation drive parked at motor $\mathrm{el}=0$ for every azimuth move (model el -3.1°, range -4.3 to 0.0°, from imu_el in the pointing table). Gravity then sits 4.4–4.6° off the shaft, so the specific force traces a small circle in imu_az's frame as the shaft turns. A sensor turned counter-clockwise about up by θ sees any fixed vector turn by $-\theta$. So the circle's phase about up changes by -1 per degree of motor az if the shaft turns counter-clockwise from above, and by $+1$ if it turns clockwise.

"Up" in imu_az's frame comes from the 2026-07-09 picohost calibration that every file carries in `header/imu_calibration`: a full elevation revolution (13 stops) with the azimuth parked. That turn tips the shaft through every direction relative to gravity, so the fitted bias ($5.1\,\mathrm{m\,s^{-2}}$) includes its component along the shaft. An azimuth turn alone cannot do this: it leaves every reading on one circle, and a bias along the shaft cannot then be told from gravity along it. The calibration still held during the test. With its bias removed the reading's magnitude is 0.992 of the calibrated g (5–95%: 0.978–1.006), and the specific force's component along the calibrated up is at least 0.969 at every settled azimuth. The shaft axis is 9.5° from the calibration's up, farther than gravity is from the shaft, so the phase is measured about the fitted circle centre. About the calibrated up itself the slope is +0.08, which is meaningless because the origin lies outside the circle.

Between the 902 settled azimuths (motor at its target for at least five samples) there are 103 steps of 1–150°: 54 with +az and 49 with −az. In 9 of them the motor count changed while the phase moved by less than 1°, so the shaft did not turn. These are count re-homes (memo 003). Of the other 94, 93 turn clockwise seen from above (Fig. 9b), with a median phase change of 0.97 per degree of motor az (5–95%: 0.33–1.20). The same code, run on a simulated sensor turned counter-clockwise, returns −1. So **+motor az turns the azimuth shaft clockwise seen from above**. This rests on four things outside this test:

- the BNO085's axes are right-handed (its datasheet, §4);

- its accelerometer reports specific force, $+g$ on the upward axis at rest;
- the firmware (`pico-firmware imu.c`) passes the axes through unchanged;
- the azimuth motor's direction did not change between these days and the 07-17 raster (Aaron, 2026-10-26).

The boresight faces up at el = 0. The HFSS pattern is not front–back symmetric, so the 07-17 transmitter raster can tell the boresight up from down. We fit it with each of the four combinations in Table 7: the az and el zeros free, the transmitter and antenna at the v0005 posterior means, on 6552 samples and 64 comb teeth. Scoring is on held-out azimuth stripes. The raster is loaded and flagged as `data-analysis/scripts/marjum-2026-07/fit_beam.py` does it, on pointing table v4, and fitted with `eigsep_data.beam_mapping` (466c0d1). With the boresight up the held-out rms is 0.097 (right-handed) and 0.088 (left-handed). With it down the fits are 2.8 times worse: 0.249 and 0.257. The beam fixes the boresight. Its preference between the handednesses, 0.008 in rms, is small beside that, and this memo does not use it; memo 003 found the polarization equally unable to choose.

Two photographs show the turn. IMG_2217 and IMG_2226, from the 91 m parallax set, show the bowtie face-on from 111 and 119 m during the 07-17 raster (Fig. 10). Between them the motor az went from -90 to -75° in steps that were all positive (table az 280.6 to 295.7°, +15°). The elevation was sweeping continuously, so each photograph is taken at the moment the pointing table's el crosses 90° (mast horizontal), the crossing nearest before its EXIF time (20:49:22.6 and 20:52:14.9 UTC). Both photographs show the mast pointing near the camera, which only happens near that crossing. The crossings fall 12 and 8 s before the EXIF times, as if the phone clock ran ahead. The 4 s difference between the two is about 20° of elevation, so the photographs fix the elevation only that well. At el = 90° the mast is 33° from the line to either camera, so both look at the dipole from in front of the mast.

Through the v0005 camera poses, with each handedness's az zero from Table 7's beam fits:

- right-handed puts the upper arm tip at +3° from the image vertical in IMG_2217 and -8° in IMG_2226;
- left-handed puts it at -6° and then +9°.

Positive angles are toward image right, which is 335° true. The two handednesses predict opposite turns. In the photographs the blades are close to vertical in IMG_2217 and the upper blade leans toward image right in IMG_2226, the left-handed turn (Aaron's reading of the photographs, checked here against the pointing). The lean looks larger than the predicted change. The elevation's uncertainty and the mast's 33° angle to the line of sight both enter, so the photographs are used for the sense of the turn only. They use the beam only for which way the mast faces, the fact the beam fixes above, and they use no IMU.

Together: the azimuth is left-handed. With the boresight up at el = 0, a clockwise turn from above is a left-handed turn about the boresight. The only combination consistent with both measurements is the left-handed azimuth with the boresight up, and the photographs show the same turn. That is the opposite of the convention in `eigsep_base` and memo 003, which this changes. Memo 003's pointing and the beam and image-domain products built on it (memos 005 and 008) still use the right-handed convention. Reversing it reverses the sign of the table's azimuth zero, to about the +16.0° that memo 003's left-handed fit gives. It also changes memo 003's reading of the transmitter's arms: under the left-handed azimuth, memo 001's label (arm 0 is rx6) stands. Memo 003's fit then puts arm 0 10.2° from the photographed rx6 axis (memo 003 used the v0004 feed axis, `tx_feed_orientation/v0002`; §10 gives v0005's), and nothing yet explains that offset (§11).

Table 7: The four combinations of azimuth sense and boresight direction at el = 0. The third column is the turn of the azimuth shaft, seen from above, that each predicts for +motor az; imu_az measures clockwise. The last two columns are the 07-17 raster fits' median normalized residual rms on the training samples and on held-out azimuth stripes (`fit_beam.py` az_el model; HFSS beam; v0005 positions).

az sense about the boresight	boresight at el = 0	shaft turn for +az, from above	imu_az (§8)	beam fit rms train	beam fit rms held out
right-handed	up	counter-clockwise	excluded	0.103	0.097
right-handed	down	clockwise	consistent	0.256	0.249
left-handed	up	clockwise	consistent	0.094	0.088
left-handed	down	counter-clockwise	excluded	0.264	0.257

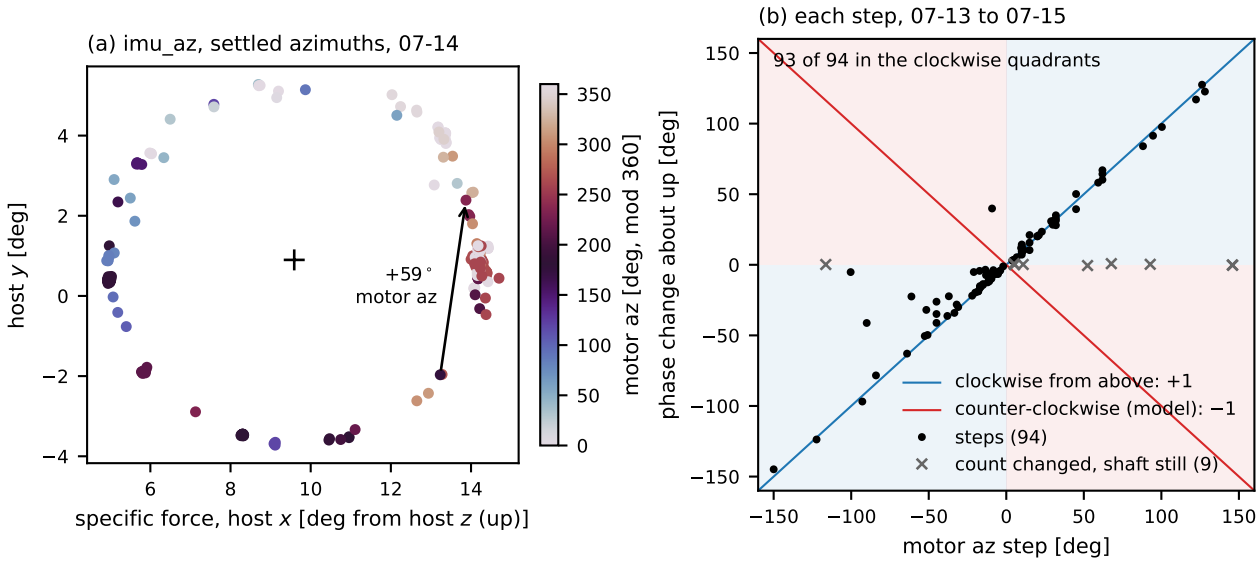


Figure 9: The azimuth shaft's turn from imu_az, read from the raw correlator files' metadata with the 2026-07-09 calibration's bias and frame. (a) The specific force at each settled azimuth on `07-14`, as angles from the calibration's up, coloured by motor az. The cross is the fitted circle centre, the shaft axis. The view is from above (host z toward the reader). The arrow is one +az step: in the sensor's frame the specific force moves counter-clockwise about the centre, so the sensor, and the shaft, turned clockwise. (b) The phase change about the circle centre against the motor az step, for every step from 07-13 to 07-15 (the centre is fitted per day). Blue quadrants: the shaft turns clockwise seen from above (left-handed); red: counter-clockwise (the right-handed model). Crosses are count changes during which the shaft did not turn.

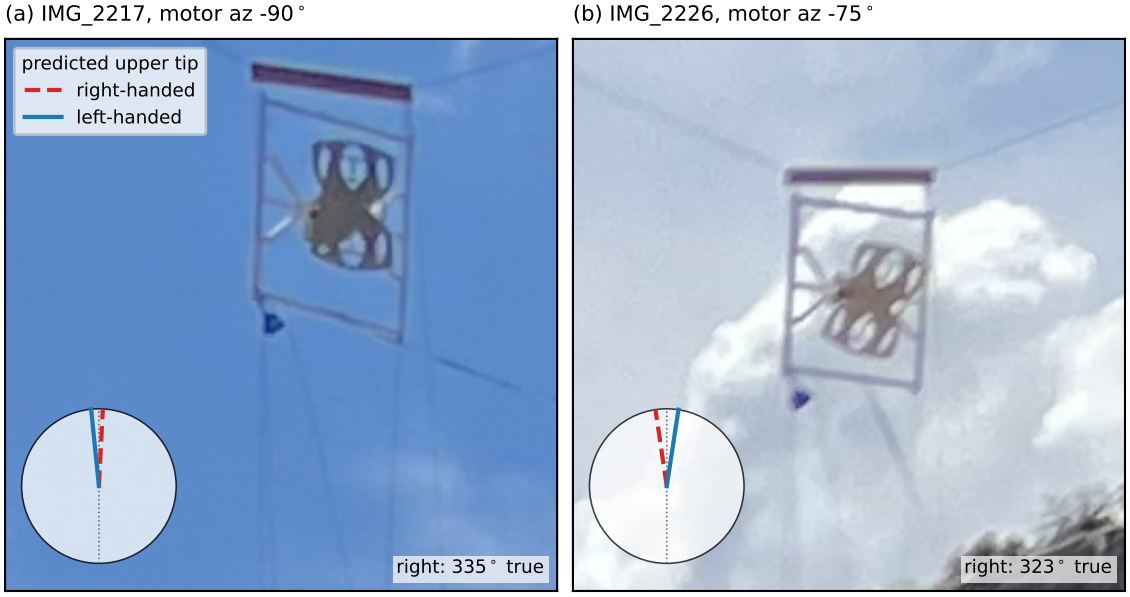


Figure 10: The dipole seen nearly end-on, from in front of the mast, before and after the +15° az step: crops of IMG_2217 and IMG_2226, centred on their labelled antenna point and upright. Each dial shows, on the same image axes, the direction of the dipole's upper tip that each handedness predicts at the el = 90° crossing (§8): v0005 camera poses and positions, with the az zeros of the beam fits in Table 7. The dotted line is image vertical. "Right" is the true bearing of image right. The predictions are model-only and are not drawn on the dipole itself.

9 Heights, height eras and the ground antenna

The 91 m-era antenna is 93.5 m above the DEM surface at its v0005 (E, N) (the historical release gave 93.4 m, as quoted in memo 001, Table 4); the LIDAR dwell reads 92.3 m, a range to whatever ground it struck; the field notes' 91 m has no recorded datum. The only height with a stated datum, 87.5 m, is "from the tether anchor", whose position is not recorded. These are different quantities and must not be averaged (`curation/height_references.json`).

9.1 Height eras from the S11 terrain echo

The in-situ VNA measures the antenna's reflection coefficient Γ_{ant}, at reference plane P, the antenna side of the RF switch. Its delay spectrum (50–250 MHz, 4-term Blackman–Harris taper) shows the echo from the terrain below. The echo's lead peak (≈569 ns on the night of 07-16/17) is the return from the nearest terrain, about 83 m away toward a true bearing of about 41°, 62° below the horizon. The gap edge before it is the peak's taper skirt, not a separate return. A DEM v0004 echo model gives the lead peak's sensitivity to antenna height, $\partial\tau/\partial U = 6.00\,\mathrm{ns\,m^{-1}}$ (5.66–6.06 across the ±3.2 m bracket of geometry v0004, on which the product was computed). Differencing the lead peak between S11 epochs and dividing by that sensitivity gives the vertical change between configurations, which needs no absolute delay (Fig. 11).

Three checks support the reading:

- **The reference plane did not move.** Γ_{ant}'s dominant early reflection, at 22.9 ns (the balun–switch coax including the balun), shifts by at most 0.07 ns between epochs.
- **The rotations do not matter.** The feed is on the azimuth axis by design, so azimuth rotations between epochs do not move it.
- **One sweep is excluded.** The 07-17 04:02 UTC sweep (22:02 MDT on 07-16) was taken while the box swung, uncommanded. The motor counts are constant, while the IMU elevation swings with a half-range of up to 2.3° from 21:54 to 22:22 MDT. That sweep changes the echo's shape and is not used.

Table 8 gives the results and their budgets:

- **The lift to the final height** (night reference against 07-17 19:10 UTC) raised the antenna by $+1.66 \pm 0.23$ m. So $U_{87.5} - U_{91} = -1.7 \pm 0.2$ m, not the -3.5 m that the field labels differ by.
- **The battery re-raise** (07-18 01:17 UTC against 07-17 19:10 UTC) left the antenna $+0.57 \pm 0.17$ m higher than the first lift. The lidar's 92.3 m was read after the re-raise and the field notes' 91 m refers to the first lift, which may account for part of the difference between them.

These are vertical-only readings. A sideways move on the leaning tethers is bounded by $|\delta U| \tan 11.4°$ and is in the budget. The changes are differences, so they hold in any frame.

The heights. The geometry photographs were taken in the first-lift configuration, so the v0005 position anchors that configuration, and the S11 changes carry the anchor to the others. The antenna stood above the DEM surface below it at:

- 91.8 m in the 87.5 m era (night of 07-16/17);
- 93.5 m after the first lift to the "91 m" configuration (the v0005 position);
- 94.0 m after the battery re-raise.

The differences between these heights are good to ± 0.2 m. All three share the ± 3.5 m bracket on the v0005 position. The ∼30 m era has no S11 epoch; its position is triangulated from photogrammetry in §9.2. The field labels (87.5 m from the tether anchor, 91 m with no datum) and the LIDAR's 92.3 m slant range measure other things and are not comparable with these heights (Fig. 11b).

The absolute echo delay needs the delay from plane P to the antenna's radiation centre, which is not known. It shows no conflict with the v0005 height, but it is not a test of it. Label: diagnostic. Product: `derived/height_eras/v0001`.

Table 8: Lead-peak vertical change δU between S11 epochs and its error budget σ, the sum in quadrature of: stat., epoch-to-night statistics; night, the night's own step (the last eight reference epochs sit 1.2 ns later than the first four); taper, taper and floor choice; plane, the reference-plane check; lean, the tether-lean bound; bracket, the ± 3.2 m position bracket of geometry v0004, on which the product was computed (v0005's ± 3.5 m would raise it by about 0.006 m; all in m). Night = the 12 S11 epochs of 07-17 04:44–15:25 UTC. Times are UTC.

epochs	δU [m]	σ [m]	stat.	night	taper	plane	lean	bracket
lift: 07-17 19:10 − night	+1.66	0.23	0.12	0.10	0.02	0.01	0.16	0.06
07-18 01:17 − night	+2.23	0.28	0.12	0.10	0.00	0.01	0.22	0.07
re-raise: 07-18 01:17 − 07-17 19:10	+0.57	0.17	0.16	0.00	0.02	0.01	0.06	0.02

9.2 The 30 m sky antenna

Box-air was lifted from the ground to ∼30 m while the recorder wrote no files, 07-15 21:29–23:11 UTC, and stayed suspended until it was lowered on 07-16 (memo 001). Two transmitter-era photographs, IMG_2171 and IMG_2172 (07-15 22:57–22:58 UTC; EXIF times are MDT), show it in that gap. Whether it had reached its resting height when they were taken is not recorded. The rays through the labelled antenna pixels, at each chain draw's poses, triangulate to a mean of (1654.90, 2035.57, 1706.76) m (UTM grid; Tables 1 and 2), 20.3 m above the DEM surface. Relative to the 91 m antenna it is +4.61 m in N, −0.23 m in E and −71.10 m in height. Two rays determine a point but leave no residual to test it, and the position moves with the transmitter-era poses (Fig. 7; $r = 0.91$ with the transmitter's E). What moved the antenna north of the 91 m position, and whether it was still rising, is not established.

What a comb delay can test. Two receivers see a comb from a source at $\mathbf{x}_s$ with a cross-correlation lag

$$\tau = \frac{|\mathbf{x}_s - \mathbf{x}_{\mathrm{air}}| - |\mathbf{x}_s - \mathbf{x}_{\mathrm{gnd}}|}{c} + \tau_{\mathrm{inst}},$$

where τ_{inst} is the difference of the two signal chains' cable and digitisation delays. τ_{inst} is fixed while the SNAP runs continuously and can change across a power cycle; it has not been measured. A single lag therefore does

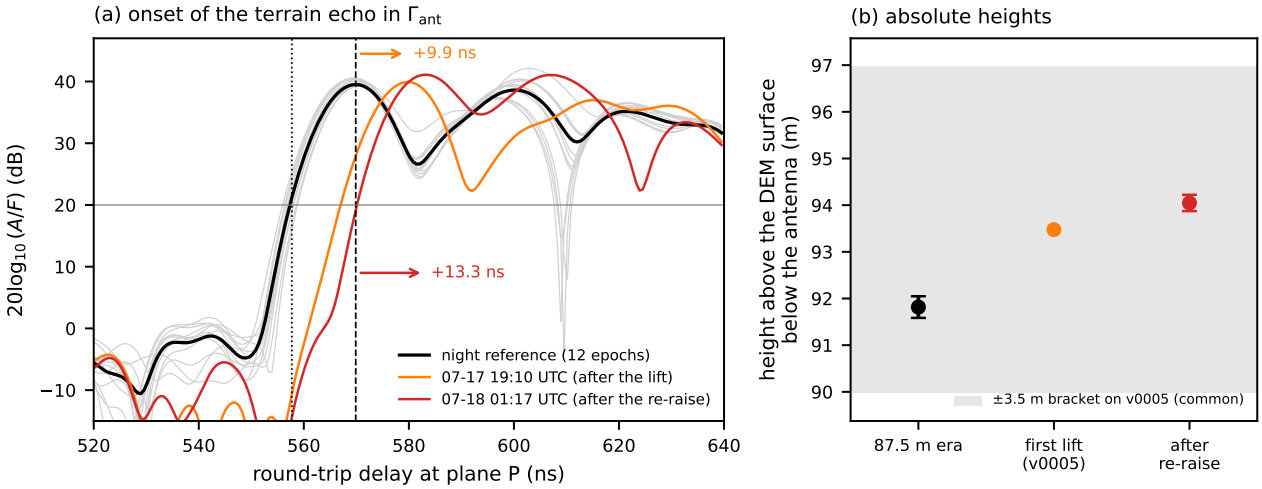


Figure 11: Height changes from the terrain echo. (a) Delay spectra of the calibrated antenna reflection Γ_{ant} at plane P, 520–640 ns, as $20\log_{10}(A/F)$ over the floor F (4-term Blackman–Harris taper). Grey: the 12 night-reference epochs (07-17 04:44–15:25 UTC); black: their power-combined spectrum. Dotted and dashed lines: the night's mean gap edge and lead peak. Coloured: the epochs after the lift to the final height and after the battery re-raise. Arrows: the shift of the lead peak, the return from the nearest terrain. (b) Absolute height above the DEM surface below the antenna for the three configurations: the 87.5 m era (the 12 night epochs, 07-17 04:44–15:25 UTC), the first lift (07-17 19:10 UTC) and after the battery re-raise (07-18 01:17 UTC). The first lift is the v0005 position. The others follow from it by δU = lead shift divided by $\partial\tau/\partial U = 6.00\,\mathrm{ns\,m^{-1}}$, with the total error of Table 8. The grey band is the ±3.5 m bracket on v0005, common to all three.

not test a position. The change in lag between two box-air positions within one continuous SNAP run does, because τ_{inst} and the box-gnd term cancel. With the transmitter as the source, the S11 lift of +1.66 m (§9) predicts a lag change of 5.5 ns across the 07-17 lift to 91 m.

The 07-17 lift in the comb lag. `validate_comb_lag.py` makes this test (Fig. 12). The SNAP did not re-sync between 15:30 and 21:30 UTC: all 167 files carry one sync time (07-07-17 03:22:45 UTC), gateware, board and configuration; the accumulation counter steps by one through every file and never resets (0 breaks); and the last sample of each file stays -0.46 to -1.46 s from the file's close time on the Panda clock. Per file, the 49 comb teeth at 140–235 MHz are averaged coherently over the integrations where the comb is on in both inputs, and the lag is the weighted least-squares slope of the phase against a reference spectrum (the mean of the 91 m parked files), with a free phase offset. Only parked files with the antenna switched in and the final drive setting (from 18:14:47 UTC) are compared.

- **The lag steps once.** It is −6.50 ns (6 files, s.d. 0.10 ns) before 18:27 UTC and −0.25 ns (9 files, s.d. 0.11 ns) from 18:29 to 18:49, a change of 6.25 ± 0.05 ns from the scatter between files. At a fixed height and pointing (91 m, 19:00–19:52, 21 files over 50 min) the lag scatters by 0.16 ns (0.73 ns peak to peak) and drifts by +0.02 ns/h, so changes below ±0.36 ns (half the largest peak-to-peak range at fixed height) are not resolved.
- **It is a path change of** 1.88 ± 0.02 ± 0.11 **m** (scatter, then fixed-height wander; projected through the 3.67° nadir angle to the transmitter), against the S11 lift of +1.66 ± 0.23 m, 0.9σ apart. The two halves of the band, below and above 188 MHz, give lags within 0.81 ns of each other in every compared file, so the change is not dispersive.
- **It happens at 18:27:28–18:28:03 UTC, not at the mode table's 18:48/18:51 boundary or the field notes' ∼18:50.** The lag ramps over 35 s, while box-air's IMU roll swings between −177° and −166° and settles at −174°, against −177° before. Box-air's comb tooth power changes by -0.24 dB and box-gnd's by -0.02 dB, so the transmitter did not change. Across the mode table's boundary and the 59° azimuth turn at 18:51 the lag moves by +0.16 ns; with the antenna at el ≈ 0, memo 003's phase-centre lever arm

contributes under 0.2 cm there. The coherence edge of the lift (`height_eras/v0001`) is good to ±20 min and does not exclude 18:28. Rows labelled ∼87.5 m from 18:28 to 18:51 were therefore taken in the 91 m configuration.

- **Caveats.** The sign assumes the correlator forms $V_{04} = \langle x_0 x_4^* \rangle$, so that a positive change is a longer path to box-air; that convention has not been checked, and only the magnitude is compared with the S11 lift. At a fixed height the lag differs by up to ∼2 ns between the transmitter drive settings of the 15:36–18:13 tests (Fig. 12a), so only files at the final setting are compared.

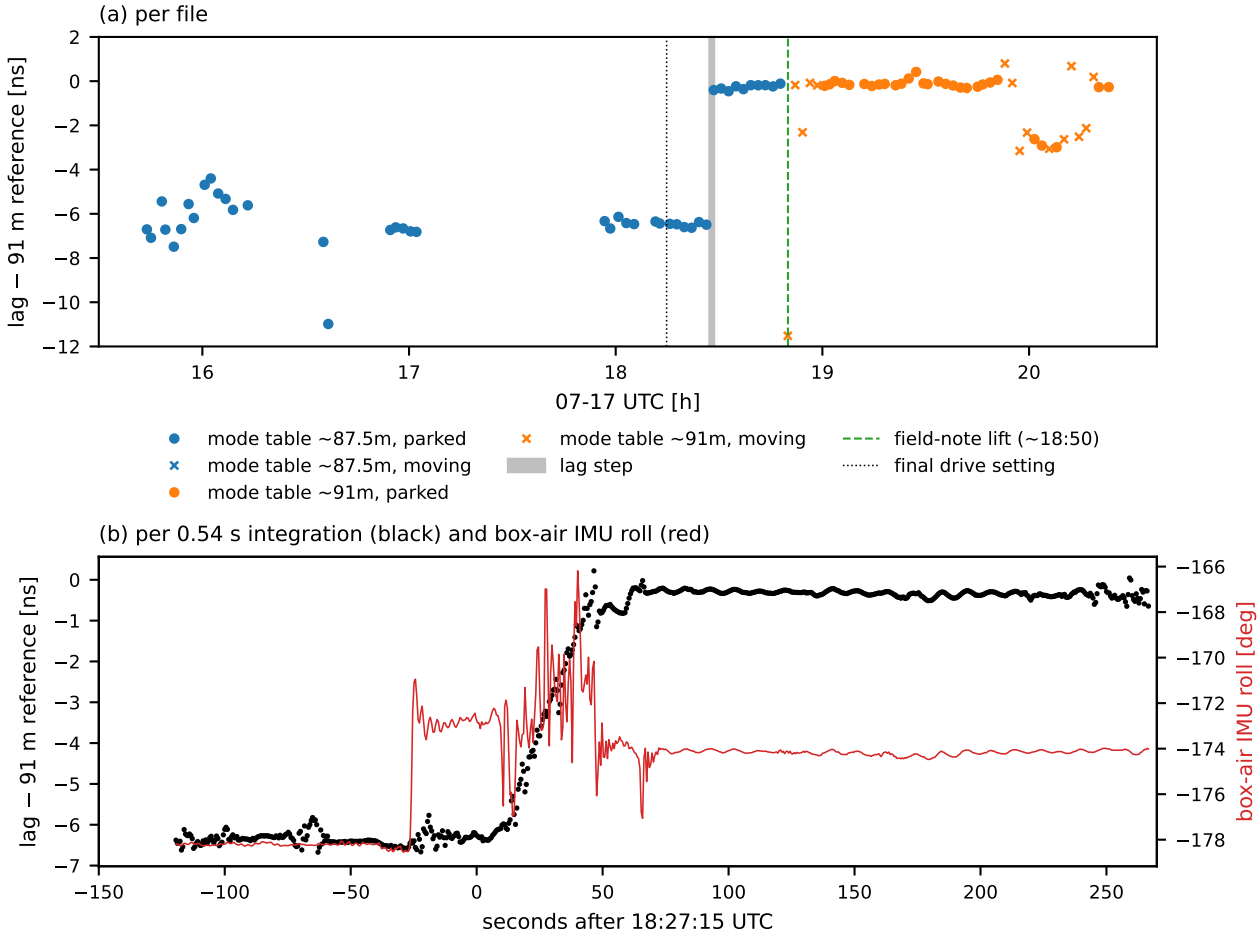


Figure 12: The transmitter comb's box-air × box-gnd lag on 07-17 (`validate_comb_lag.py`, raw correlator files, teeth 140–235 MHz), relative to the mean spectrum of the 91 m parked files. (a) Per file, coloured by the height era of `curation/mode_table.jsonl` (circles parked, crosses moving); grey, the step; dashed, the field-note lift time; dotted, the start of the final drive setting. The scatter before 18:14 is the drive tests. (b) Per 0.54 s integration through the step, with box-air's IMU roll (red). The lag moves by 6.25 ns in 35 s while the box swings, ∼20 min before the recorded lift.

An earlier version of this section matched a 287.1 ns lag "on 07-16 02:40 UTC, during the transmitter comb run" to an 86.3 m distance between the 30 m antenna and the transmitter. That check is withdrawn. No code or product records the lag; the transmitter comb first appears on 07-17 at 15:36 UTC, and the comb on box-air in the early hours of 07-16 is box-air's own 1.000 MHz emission (memo 001), whose source is box-air itself; the 30 m antenna is 24.1 m from the transmitter, not 86.3 m; and a lag compared with one distance ignores τ_{inst}.

9.3 The ground antenna (`box-gnd`)

`box-gnd` is the second receiver's dipole, mounted on its box on the ground near the transmitter. Six photographs show it (Table 9): IMG_2157 (07-15) and IMG_2187, 2189, 2191, 2194 and 2246 (07-17). None is one of the v0005 cameras. The box did not move between them (Aaron). By IMG_2246 (21:24 MDT) the feed had tipped over, in the wind storm that began about 21:00 MDT on 07-17 and brought box-air down about five minutes later (Aaron), so that frame shows the box but not the dipole.

Poses. The 29 v0005 cameras are held at their release poses, and the six photographs are solved against them with `validate_box_gnd.py`, together with IMG_2204, a Theodolite-app screenshot that tests the app's readings (*Dipole orientation*). Appendix C shows each of these seven fits: tie points and their residuals, the skyline against the DEM horizon, and the picks against the fitted points. Tie points are RootSIFT features on quarter-resolution, sky-masked images, matched between each new photograph and every other image; a pair is kept with at least 30 fundamental-matrix inliers (unrelated pairs give 11–20), leaving 25 pairs and 4283 tracks. Photographs with sky also fit their segmented skyline (`derived/image_segmentations/v0002`, made for this section with the v0001 segmenter settings) against the DEM horizon. The priors are those of the v0005 model (EXIF GPS with the v0005 common GPS bias, the EXIF heading) plus the camera 1.5 ± 2 m above the DEM; IMG_2204 takes the GPS fix printed in its overlay and no heading prior. Before any pick is used, these terms fix IMG_2191, 2194 and 2246 to within 0.11, 0.14 and 0.08 m (largest axis): they share hundreds of tie points with release cameras. IMG_2157 shares its tie points only with IMG_2194, through cliffs hundreds of metres away, which fix its orientation but leave its position along its line of sight formally free by 0.52 m; that sd is optimistic (*Checks*). IMG_2187 and 2189, taken 22 s apart, share tie points only with each other, so GPS, the skyline and the height prior alone place them (1.24 m).

Position. Aaron picked the centre of the box (the reference point of the box-air antenna labels) and the two tips of the pink-flagged dipole in each photograph (`box_gnd_picks.json`). The picks enter the same least-squares adjustment as landmarks. Only the 3 photographs whose position is known to better than 0.5 m fix `box-gnd` (IMG_2191/2194/2246); the other three are checks. The box centre is at (1654.5, 2037.6, 1687.8) m, 0.3 m above the DEM, offset (+3.9, +10.5, +3.1) m from the v0005 transmitter (Fig. 14). The picks miss the fitted point by 6–26 px, against the 3 px assumed; the uncertainty is scaled by their reduced χ^2 (28, an effective 16 px per pick) and combined with 40 posterior draws of the v0005 cameras, each re-solved. The 95% intervals are $E \in [1653.4, 1654.6]$, $N \in [2036.7, 2038.5]$ and $U \in [1687.1, 1688.3]$ m. Leaving one photograph out moves the box 2.2 m without IMG_2191, 0.5 m without IMG_2194 and 0.9 m without IMG_2246. IMG_2191, looking from the north, carries most of the constraint: IMG_2194 and IMG_2246 look at the box from nearly opposite sides, so the two of them fix it poorly along that line. The Theodolite-app fix in IMG_2205, taken beside the box, is 9.5 m from the fitted position once the v0005 GPS bias is removed; it is a check on the fit, not an input.

Checks. None of the three check photographs tests `box-gnd`; each fails on its own pose. IMG_2157's pick-free pose puts the camera 1.3 m from the fitted box, at right angles to the ray through its box-centre pick, while the photograph shows the box several metres in front of the camera (Fig. 14a). Its position is wrong by several metres, not by the formal 0.52 m. The rays through the box-centre picks of IMG_2187 and 2189 both pass 9.3 m from the fitted box, 8.7 m of it vertically, 7.5 times their formal position uncertainty. Their height comes only from the 1.5 m prior at a GPS position good to ±10 m on a slope, which the formal uncertainty does not capture; the miss is consistent with an error of that size in their height.

Dipole orientation. IMG_2205 (07-17 12:34, before the storm; Fig. 13) is a Theodolite-app screenshot taken with the phone sighted along the pink-flagged dipole: the crosshair column runs through the dipole's near tip and its centre (Aaron; the pink flag hangs just off the tip, so the sighting is on the tip, not the flag). The app reads 134° TRUE, so the pink end lies at 314° true from the centre (axis 134°, mod 180). Two errors enter. The compass: IMG_2204, the same app on the same phone 2.5 h earlier, looking down on the transmitter feed, reads 174° TRUE, and its tie-point pose (389 tie points, Fig. 25) puts the boresight at $175.3° \pm 0.6°$, a difference of $-1.3°$. We take the quadrature sum of the difference and the pose's sd, 1.5°, as the compass error. It is one comparison, at another spot; the magnetic environment beside `box-gnd` is not tested. The screenshot's tie residuals (6.5 px rms) are larger than the photographs', and include a streak along the repeated cage bars. Its fitted focal length, 965 px, is the 0.5× ultrawide preview's; refitting with the 1× lens's distortion or none

moves the boresight by at most 0.04°. The app's two other angles are not reproduced: the pose's elevation is −29.1° against the app's −25.5°, and its roll 1.7° against +4.9°. We do not know how the app defines them, and use only the heading. The alignment: a tip or centre 5 px off the column, at about 0.6 m (from the crossed arm's apparent length), turns the axis by 0.3°. **The pink end is at 314° true, ±1.5°** (95%: [311°, 317°]). IMG_2205 cannot be posed itself: it shares no view with any other image (11–20 matches per pair at quarter, half and full resolution, the level of unrelated pairs). The photogrammetric tips agree but barely constrain it: picked in IMG_2191 and 2194 they give 305°, 9° away, with a 95% interval of ±60° (IMG_2157's tip picks are too close to its weakly placed camera to use, and IMG_2246 shows the feed fallen).



Figure 13: IMG_2205, the Theodolite-app screenshot sighted along `box-gnd`'s pink-flagged dipole. (a) The whole screen: the app reads 134° TRUE at the crosshair. (b) The crosshair column, through the near tip (pink flag just beside it) and the dipole centre above the box.

Table 9: The `box-gnd` photographs and the IMG_2204 screenshot at their solved poses (`box_gnd_fit.json`). Local time is EXIF (UTC−6). Ties: tie observations kept, with the images they link to. Skyline: median |segmented − DEM horizon|. Position sd: largest axis, from tie points, skyline and priors before any pick is used. Last column: whether the photograph's picks fix `box-gnd`, and the residual of its box-centre pick.

image	local time	lens (35 mm eq.)	ties	tie rms [px]	skyline [px]	position sd [m]	box centre
IMG_2157	07-15 16:48	wide (26)	43 (2194, 2198, 2213)	0.7	—	0.52	check
IMG_2187	07-17 09:23	ultrawide (14)	1790 (2189)	1.4	6.0	1.23	check
IMG_2189	07-17 09:24	ultrawide (21)	1788 (2187)	1.0	—	1.24	check
IMG_2191	07-17 09:27	wide (31)	454 (2199, 2194, 2198)	2.7	4.1	0.11	fixes; residual 6 px
IMG_2194	07-17 09:30	wide (26)	1252 (2198, 2217, 2191)	4.2	4.1	0.14	fixes; residual 20 px
IMG_2204	07-17 10:06	screenshot (0.5×)	389 (2203, 2199, 2159)	6.5	—	0.04	compass check
IMG_2246	07-17 21:24	wide (26)	366 (2210, 2209, 2171)	2.3	3.9	0.08	fixes; residual 26 px

10 Transmitter antenna orientation

The transmitter antenna is a pair of crossed dipoles on a square feed; alternate comb teeth come from alternate dipoles, rx6 (arm 0) and rx1 (arm 1) (memo 001 §2). Its orientation is measured from IMG_2203, a transmitter-era frame taken about 2 m above the feed, looking down on it (Fig. 15). The four outer corners of the feed face, picked by Aaron, are fitted as a planar square seen through the camera's fitted pose, with its perspective and radial distortion and no assumed size. The rx6 axis joins the two quadrants marked "6" in the field; rx1 joins the other two.

Bearings here are true bearings. The poses live on the UTM raster grid, so an axis computed on it is a grid bearing; true = grid − 1.52° at the transmitter. At the v0005 pose of IMG_2203 the square fits with a corner rms of 1.6 px and a tilt of 0.1° from horizontal, and **the rx6 axis lies at 165.1°, rx1 at 75.1°** (mod 180).

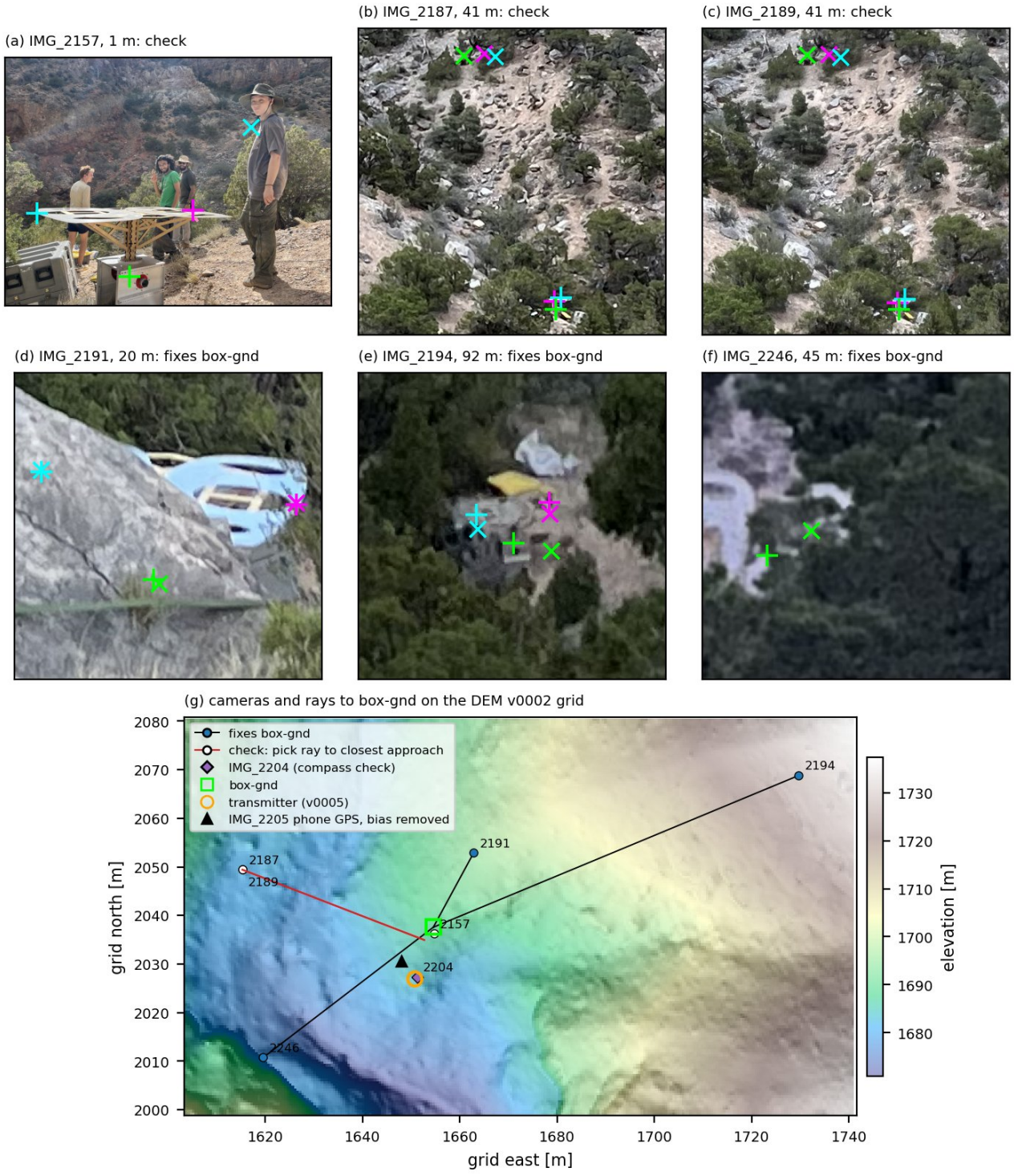


Figure 14: `box-gnd` in its six photographs and on the map. (a)–(f) Crops around the box: Aaron's picks (+) of the box centre (green) and the pink-flagged (magenta) and opposite (cyan) dipole tips, and the fitted points projected through each photograph's solved pose (×, same colours). The title gives the range and whether the photograph fixes the position or is a check. (g) The cameras on the DEM v0002 grid, with lines to the fitted box from the photographs that fix it (black) and, for the checks, the ray through the box-centre pick to its closest approach to the box (red); the v0005 transmitter (orange) and the IMG_2205 phone fix with the GPS bias removed (triangle).

A second planar solution, nearly vertical, fits worse and is rejected because the feed sits flat in its dish. Across 500 draws of the v0005 chains the rx6 axis is 165.6° with sd 0.83°. Moving the corners by 2 px, about their rms, spreads it by 0.39°, and letting the face be 10% off square moves it over 164.9–165.4°. Against the historical solution (`tx_feed_orientation/v0001`, at the historical pose and MCMC draws), the axis moves −0.22° on the grid; the larger change is the frame: the historical 167.3° was a grid bearing reported as true.

Adopted: rx6 axis 165.6° ± 1° true (NNW–SSE), rx1 axis 76° (ENE–WSW).

The theodolite-app reading in IMG_2204, corrected to the arm as in the legacy geometry memo, is 166.71°, 1.1° from this axis. The screenshot shows the app set to TRUE north, and its tie-point pose (§9.3) puts the crosshair at 175.3° true against the app's 174°, so the app's heading there is good to about a degree; the correction from the crosshair to the arm is the legacy memo's and is not redone here. IMG_2202 agrees once it is read correctly. It was taken at the transmitter, looking across the feed, and its horizon fit gives a camera heading of 131.3° true; an independent match of 20 canyon features gives 131.9°, so the heading is sound. The legacy geometry memo read that heading as a polarization axis, which it is not: the clear zip ties that attach the feed's corners to its support sit to the left and right of the sightline, so the camera looks across a flat edge of the square, about 45° from both arms. The feed also sits left of the image centre, so the camera to feed bearing is about 124–125°. That places the arms near 79–80° and 169–170°, within 4–5° of the IMG_2203 measurement, as expected for a sighting by eye. Comparing this with the polarization angle fitted in the beam maps belongs to the beam memo. The analysis is `scripts/marjum-2026-07/geometry/tx_feed_orientation.py` in `data-analysis` (commit 127d56c), which reproduces the historical product from its frozen inputs; its output is `marjum-2026-07/derived/tx_feed_orientation/v0003` (Appendix D).

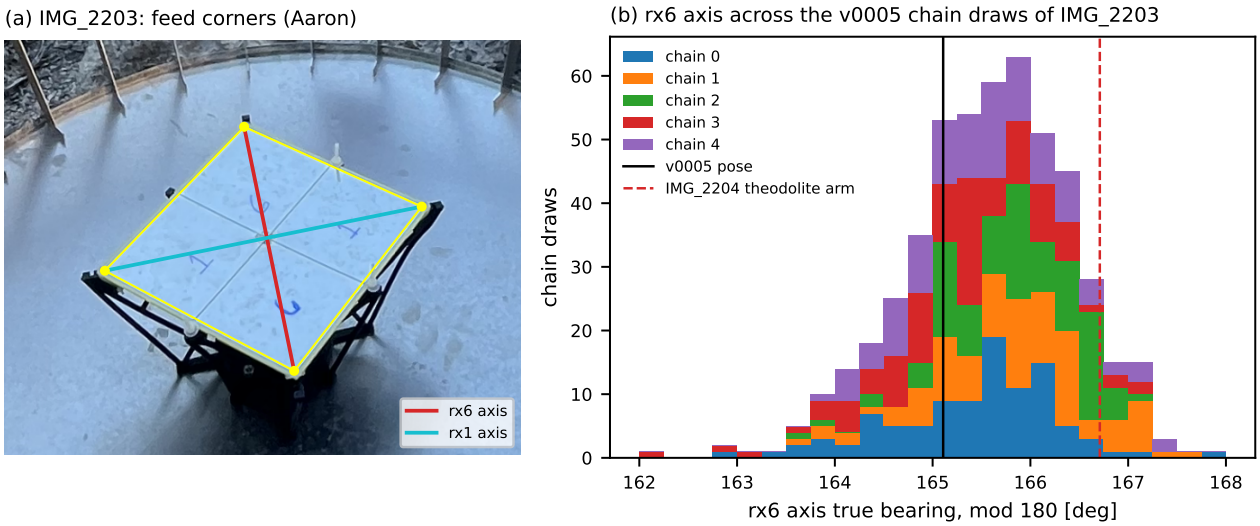


Figure 15: (a) The transmitter feed in IMG_2203 with Aaron's corner picks; the rx6 axis joins the quadrants marked "6", rx1 the other two. (b) The rx6 axis bearing solved through each of 500 draws of the v0005 chains for IMG_2203, stacked by chain, as a true bearing; the solid line is the v0005 release pose, dashed the IMG_2204 theodolite-app arm (app set to TRUE north; §9.3).

11 Open questions

- Running the v0005 model to convergence. At 2,000 draws per chain 210 of 213 coordinates fail $\hat{R} \leq 1.01$, worst for focal lengths and tilts (§5). Until a run passes, the posterior intervals are approximate.
- The 1.7σ tension between the LIDAR and GPS az-zero on the UTM grid, and the DEM's untested horizontal survey offset.
- The tether anchor's position, needed to put the field heights on a datum.
- A bench test of the BNO085's frame and sign: rotate a spare unit by hand, counter-clockwise seen from

above, through the same firmware, and check that the azimuth-handedness code returns -1 (§8). It tests the two vendor conventions the left-handed result rests on.

- Under the left-handed azimuth, memo 003's polarization fit puts transmitter arm 0 10.2° from the photographed rx6 axis (§8).
- The correlator's conjugation convention for V_{04}, which fixes the sign of the comb-lag change (§9.2).
- Whether the antenna held its horizontal position across lifts (no parallax frames outside the 91 m era). The S11 height changes assume vertical moves, with a sideways move bounded by the tether lean.
- A photogrammetric check of the lift. IMG_2198 (07-17 15:32 UTC, in the 87.5 m configuration) was taken from the canyon floor and shows the suspended antenna, which is not labelled. Its pose is usable but transmitter-conditioned (Appendix A), and it is one of the two 14 mm ultrawide frames whose distortion model remains suspect. Labelling the antenna gives one ray. At the v0005 (E, N) that ray's elevation gives the 87.5 m-era height, independent of the S11 echo. The S11 lift of +1.66 m and the field labels' 3.5 m differ by almost 2 m. It is one frame, so it carries that pose's error and assumes no horizontal move.
- The LIDAR dwell's lever arm, to check. The dwell's residual, measured minus the range predicted from the v0005 antenna, is -1.7 m (§7). The LIDAR sits on the box, near the drive, while v0005 and the S11 heights refer to the dipole, which sits above the drive by about the antenna coax length (field account). The dwell was also taken after the battery re-raise, which the S11 echo puts +0.57 m above the first lift (§9). Modelling the vertical offset from the LIDAR's mount to the labelled antenna point, in the dwell orientation, would test whether the residual is that offset. If it is, the dwell becomes an independent absolute height for the post-re-raise configuration, which the S11 echo cannot give.

Revision history

2026-10-26 New §8: the azimuth is left-handed about the boresight, against the right-handed convention of memo 003. imu_az shows that the shaft turns clockwise from above for +motor az, the beam shows that the boresight faces up at el = 0, and IMG_2217/2226 show the turn.

2026-10-26 The `v0005` chains are not converged (max $\hat{R}$ 1.36, minimum bulk ESS 11.8); the statements that they were are withdrawn, and the posterior intervals are marked approximate. The log-density gain quoted over v0004 is dropped: the two runs fit different tie-point sets.

2026-10-24 The ground antenna is now measured: six photographs of `box-gnd` are posed against the v0005 cameras and its box centre and pink-flagged dipole are fitted from picks in them (§9.3). It replaces the earlier position, the IMG_2205 phone GPS fix with an interval taken from the transmitter's. The dipole bearing is still IMG_2205's 134° sighting, now with the pink end identified and an error derived from a tie-point test of the app's compass (IMG_2204), replacing the asserted ±3°.

2026-10-23 The transmitter comb's lag measures the 07-17 lift: 6.25 ns, a path change of 1.88 m against the S11 lift of +1.66 m, at 18:27:28–18:28:03 UTC, about 20 min before the mode table's boundary (§9.2).

2026-10-23 Adopted positions are now posterior means over the chain draws, not the selected highest-density state, which no draw reaches; they move by 0.24 m (antenna) and 0.54 m (transmitter). Added the corner plot (Fig. 7). Withdrew the 287.1 ns comb-delay check of the 30 m antenna: no transmitter ran on 07-16, the quoted 86.3 m matches no distance in the geometry, and a single lag carries an unmeasured instrumental delay (§9.2).

2026-10-23 Key geometry parameters table (Table 1) added to the Summary, reporting adopted coordinates and marginalized 95% credible intervals for sky (91 m and 30 m), ground, and transmitter antennas, the Δaz = 0 boresight zero orientation, and transmitter and ground antenna dipole alignments. Added photogrammetric triangulation of the 30 m sky antenna from IMG_2171/2172 (§9.2) verifying catenary sag (+4.26 m North) and comb delay, and documented the equipment bench ground receiver setup (`box-gnd`; §9.3).

2026-10-22 Heights: the S11 terrain echo measures the height change between configurations. The lift to the final height raised the antenna by +1.66 m, so $U_{87.5} - U_{91} = -1.7 \pm 0.2$ m, not -3.5 m. The battery re-raise left it +0.57 m higher than the first lift (product `height_eras/v0001`).

2026-10-17 Adopted geometry `v0005`: full multi-view network linkage across all 29 usable cameras (3,993 landmarks), volume-preserving HMC (median $\hat{R} = 1.05$, mode log-density -1983.02), and tightened held-out brackets (3.5 m antenna, 0.1 m transmitter). LIDAR constraint v0003 brings the az-zero tension down to 1.7σ.

2026-10-10 The MCMC section, its figures and the chain rows of Table 2 now come from the v0004 chains, not the historical run, which had been plotted against v0004. They show v0004 to be a higher-density state than any chain reaches, and they overturn the historical finding that cameras without tie points split into separate modes: most separating cameras have tie points. The feed orientation is re-solved at the v0004 pose and draws and given as a true bearing (165.6°); the earlier 167.3° was a grid bearing.

2026-10-09 Held-out refits and a UTM-grid LIDAR fit give v0004 bounds of 3.5 m (antenna) and 0.1 m (transmitter); the LIDAR–GPS az-zero tension rises to 1.7σ once the LIDAR bearings are applied on the grid.

2026-10-08 Adopted geometry `v0004` as the release: positions, figures, ray misses and horizon checks are now at the v0004 poses on the UTM raster grid, with no uncertainty established; the `v0001` positions and bounds are historical.

2026-10-07 Corrected the historical tangent-versus-UTM frame description and added the v0004 UTM-target run, its nonconvergence, the carried point-fit selection and the HEIC GPS comparison.

2026-10-07 Moved the October camera proposal study into this memo, distinguished the corrected v0003 provisional point fit from a posterior, and migrated the memo's historical terrain inputs into a versioned campaign product.

2026-09-30 First version, graduating the MCMC review, geometry explorer and LIDAR explorer notebooks (`data-analysis/notebooks/arp/marjum-2026-07`) and the geometry parts of legacy MEMO-012. Same day: the transmitter antenna orientation (§10), withdrawing MEMO-012's reading of IMG_2202's camera heading as a polarization axis; and per-image solutions (Appendix B).

A Image inventory

From release `v0005` (`imgs/fits/v0005_marjum_geometry/cameras.jsonl`), which carries the geometry and labels the v0005 model used. EXIF times from the photographs. "TX-cond.": fitted with the transmitter label and no independent tie to the antenna; labels: A antenna, T transmitter.

image	EXIF (MDT)	group	pose	lens	f [px]	labels	exclusion reason
IMG_2159	07-15 16:49:12	tx-era	usable (TX-cond.)	0	2992	T	
IMG_2171	07-15 16:57:26	tx-era	usable (TX-cond.)	0	4503	T+A30	
IMG_2172	07-15 16:58:00	tx-era	usable (TX-cond.)	0	4488	T+A30	
IMG_2198	07-17 09:32:34	tx-era	usable (TX-cond.)	1	1592	T	
IMG_2199	07-17 09:36:14	tx-era	usable (TX-cond.)	0	3001	T	
IMG_2203	07-17 09:05:34	tx-era	usable (TX-cond.)	1	1640	T	
IMG_2209	07-17 14:42:18	91 m parallax	usable	1	2348	A	
IMG_2210	07-17 14:42:33	91 m parallax	usable	1	2726	A+T	
IMG_2211	07-17 14:43:29	91 m parallax	usable	1	2415	A+T	
IMG_2213	07-17 14:47:30	91 m parallax	usable	0	3194	A	
IMG_2214	07-17 14:49:05	91 m parallax	usable	0	6471	A	
IMG_2215	07-17 14:49:07	91 m parallax	usable	0	6424	A	
IMG_2216	07-17 14:49:14	91 m parallax	usable	1	2562	A	
IMG_2217	07-17 14:49:22	91 m parallax	usable	0	3916	A	
IMG_2218	07-17 14:49:28	91 m parallax	usable	0	8358	A	
IMG_2219	07-17 14:49:40	91 m parallax	excluded	—	—	A	Borderline registration: 67.9 px raster-horizon RMS and sky NLL 9.56; not promoted
IMG_2220	07-17 14:50:48	91 m parallax	usable	0	5432	A	
IMG_2221	07-17 14:51:05	91 m parallax	usable	0	3342	A	
IMG_2222	07-17 14:51:11	91 m parallax	usable	0	4637	A	
IMG_2223	07-17 14:51:13	91 m parallax	usable	0	4651	A	

image	EXIF (MDT)	group	pose	lens	f [px]	labels	exclusion reason
IMG_2224	07-17 14:51:22	91 m parallax	usable	0	6612	A	
IMG_2225	07-17 14:52:02	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2226	07-17 14:52:14	91 m parallax	usable	0	3876	A	
IMG_2227	07-17 14:52:44	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2228	07-17 14:52:49	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2230	07-17 14:56:02	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2231	07-17 14:56:47	91 m parallax	usable	0	7793	A	
IMG_2232	07-17 14:57:03	91 m parallax	usable	1	1937	A	
IMG_2234	07-17 14:58:12	91 m parallax	usable	0	3794	A	
IMG_2235	07-17 14:59:46	91 m parallax	usable	1	2943	A	
IMG_2237	07-17 15:03:21	91 m parallax	usable	0	9107	A	
IMG_2238	07-17 15:03:43	91 m parallax	usable	1	2883	A	
IMG_2239	07-17 15:04:30	91 m parallax	usable	0	3648	A	
IMG_2241	07-17 15:05:22	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2242	07-17 15:06:34	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2243	07-17 15:07:25	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2245	07-17 15:10:03	91 m parallax	usable	0	5315	A	

B Per-image solutions

One panel per usable image, at the v0005 poses. Green: the sky/ground boundary segmented from the photograph (`derived/geometry_memo_inputs/v0001/img_seg_IMG_*.npz`); red: the DEM horizon projected through the fitted pose. Plus signs mark labelled image pixels: 91 m sky antenna (yellow), 30 m sky antenna (magenta; IMG_2171/2172), and transmitter (orange). Crosses are the corresponding model positions projected through each camera pose (cyan). Each title gives the median horizon offset and, for each label, the offset in pixels and the perpendicular miss of the labelled ray in metres. The table repeats these, with the distance to the antenna. The "horizon" offset is the median over image columns where both boundaries exist and the ground below the segmented boundary is not tree (trees hide the terrain skyline). Pixel offsets scale inversely with range: the transmitter-era cameras were 2–30 m from the transmitter, so a 0.5 m miss can be hundreds of pixels.

				antenna			transmitter	
image	group	f [px]	horizon [px]	range [m]	miss [px]	miss [m]	miss [px]	miss [m]
IMG_2159	tx-era	2992	5	—	—	—	0	0.00
IMG_2171	tx-era	4503	6	—	—	—	0	0.00
IMG_2172	tx-era	4488	6	—	—	—	1	0.01
IMG_2198	tx-era	1592	3	—	—	—	0	0.00
IMG_2199	tx-era	3001	4	—	—	—	0	0.00
IMG_2203	tx-era	1640	8	—	—	—	0	0.00
IMG_2209	91 m parallax	2348	3	105	13	0.47	—	—
IMG_2210	91 m parallax	2726	3	105	12	0.40	1	0.01

				antenna			transmitter	
image	group	f [px]	horizon [px]	range [m]	miss [px]	miss [m]	miss [px]	miss [m]
IMG_2211	91 m parallax	2415	3	103	17	0.53	1	0.01
IMG_2213	91 m parallax	3194	5	85	6	0.13	—	—
IMG_2214	91 m parallax	6471	8	110	32	0.54	—	—
IMG_2215	91 m parallax	6424	7	108	54	0.88	—	—
IMG_2216	91 m parallax	2562	4	111	7	0.28	—	—
IMG_2217	91 m parallax	3916	8	111	52	1.22	—	—
IMG_2218	91 m parallax	8358	10	110	9	0.12	—	—
IMG_2220	91 m parallax	5432	10	120	4	0.09	—	—
IMG_2221	91 m parallax	3342	4	118	19	0.60	—	—
IMG_2222	91 m parallax	4637	6	118	2	0.05	—	—
IMG_2223	91 m parallax	4651	7	119	9	0.19	—	—
IMG_2224	91 m parallax	6612	10	118	31	0.54	—	—
IMG_2226	91 m parallax	3876	7	119	9	0.22	—	—
IMG_2231	91 m parallax	7793	11	124	88	1.34	—	—
IMG_2232	91 m parallax	1937	4	123	16	0.99	—	—
IMG_2234	91 m parallax	3794	5	122	8	0.26	—	—
IMG_2235	91 m parallax	2943	6	118	97	3.29	—	—
IMG_2237	91 m parallax	9107	14	94	7	0.07	—	—
IMG_2238	91 m parallax	2883	5	94	33	0.95	—	—
IMG_2239	91 m parallax	3648	5	101	65	1.67	—	—
IMG_2245	91 m parallax	5315	6	160	17	0.51	—	—

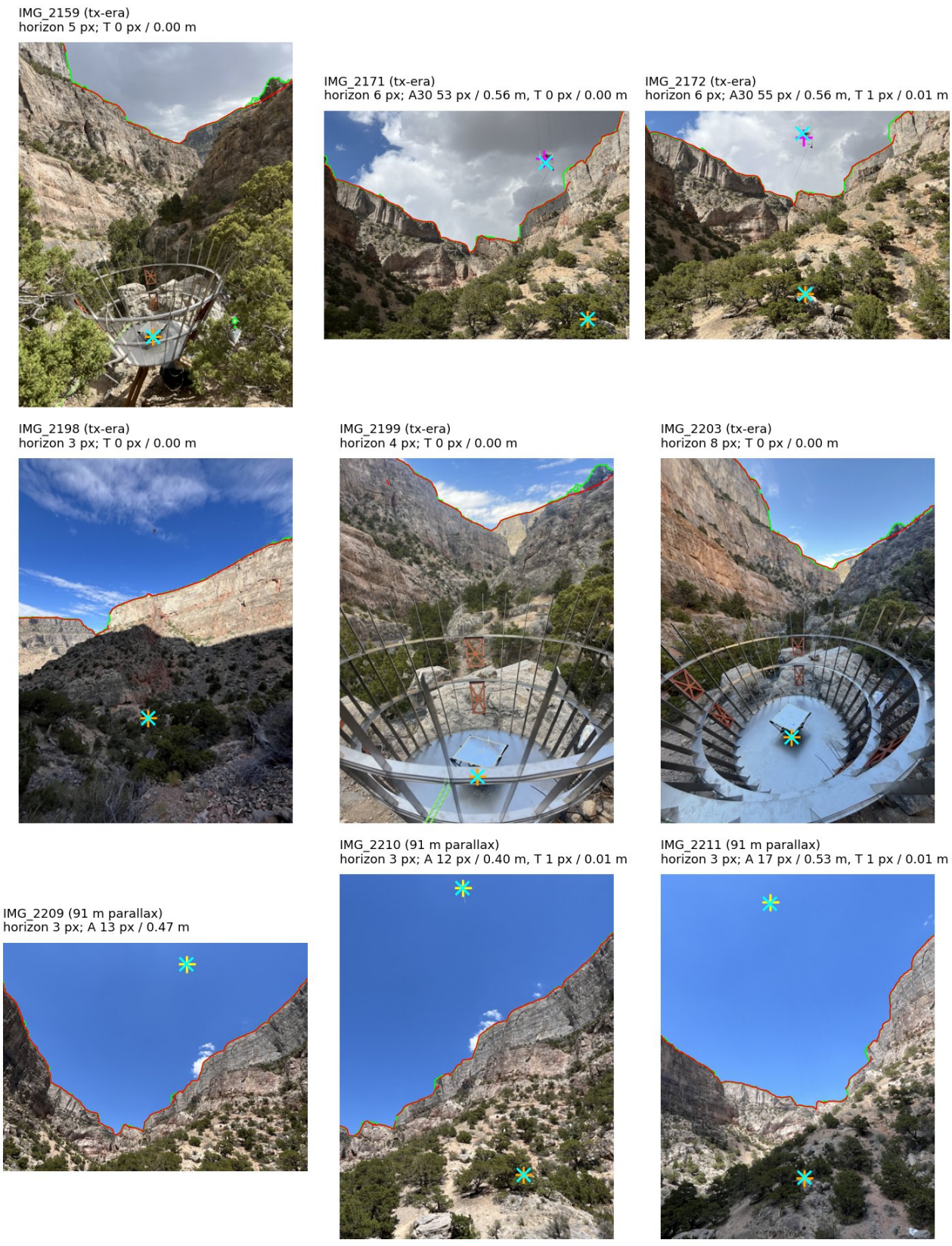


Figure 16: Per-image solutions, page 1 of 4; see Appendix B.

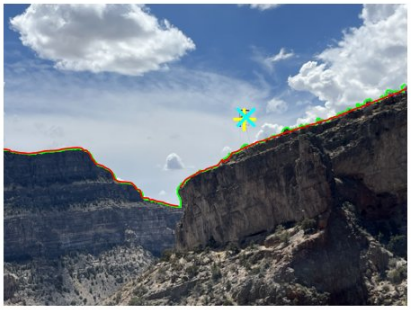


Figure 17: Per-image solutions, page 2 of 4; see Appendix B.

IMG_2223 (91 m parallax)
horizon 7 px; A 9 px / 0.19 m



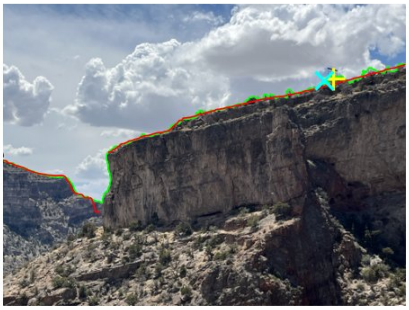
IMG_2224 (91 m parallax)
horizon 10 px; A 31 px / 0.54 m



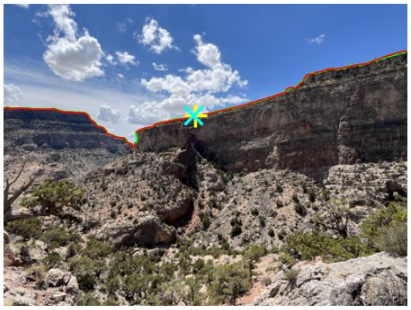
IMG_2226 (91 m parallax)
horizon 7 px; A 9 px / 0.22 m



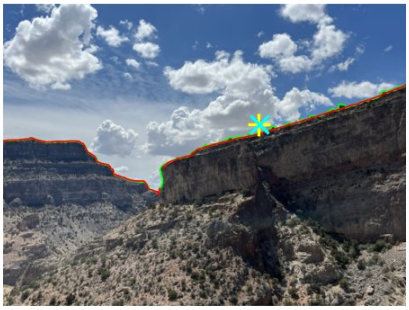
IMG_2231 (91 m parallax)
horizon 11 px; A 88 px / 1.34 m



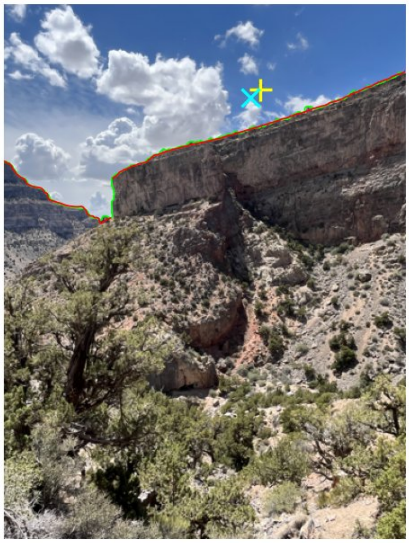
IMG_2232 (91 m parallax)
horizon 4 px; A 16 px / 0.99 m



IMG_2234 (91 m parallax)
horizon 5 px; A 8 px / 0.26 m



IMG_2235 (91 m parallax)
horizon 6 px; A 97 px / 3.29 m



IMG_2237 (91 m parallax)
horizon 14 px; A 7 px / 0.07 m



IMG_2238 (91 m parallax)
horizon 5 px; A 33 px / 0.95 m

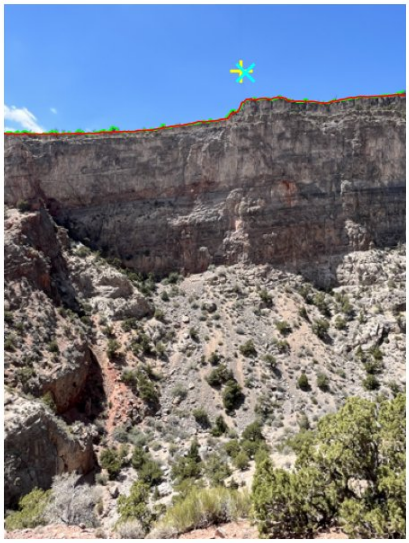


Figure 18: Per-image solutions, page 3 of 4; see Appendix B.

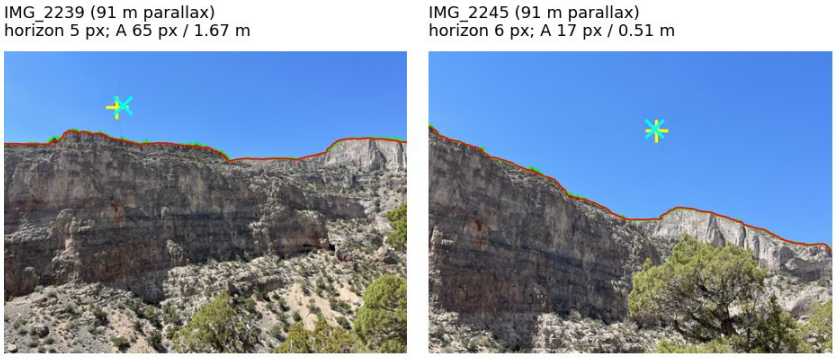


Figure 19: Per-image solutions, page 4 of 4; see Appendix B.

C Fits of the `box-gnd` photographs

One figure per photograph solved in §9.3, at its fitted pose (`box_gnd_diag.npz`). (a) The photograph: the sky boundary segmented from it (green, `derived/image_segmentations/v0002`), the DEM horizon projected through the pose (red), the skyline samples the fit uses (cyan), and the tie points kept (coloured by their position across the image) or dropped as outliers (grey ×). (b) The image it shares most tie points with, showing the same points in the same colours: matching colours in matching places are what make them tie points. (c) The tie residuals, projected minus observed, with the 4 px tie σ dashed. (d) The segmented skyline minus the DEM horizon, per image column, where the photograph has usable sky. (e) Aaron's picks (+) against the fitted `box-gnd` points projected through the pose (×; green box centre, magenta pink tip, cyan opposite tip); for IMG_2204, the crosshair the app reads.

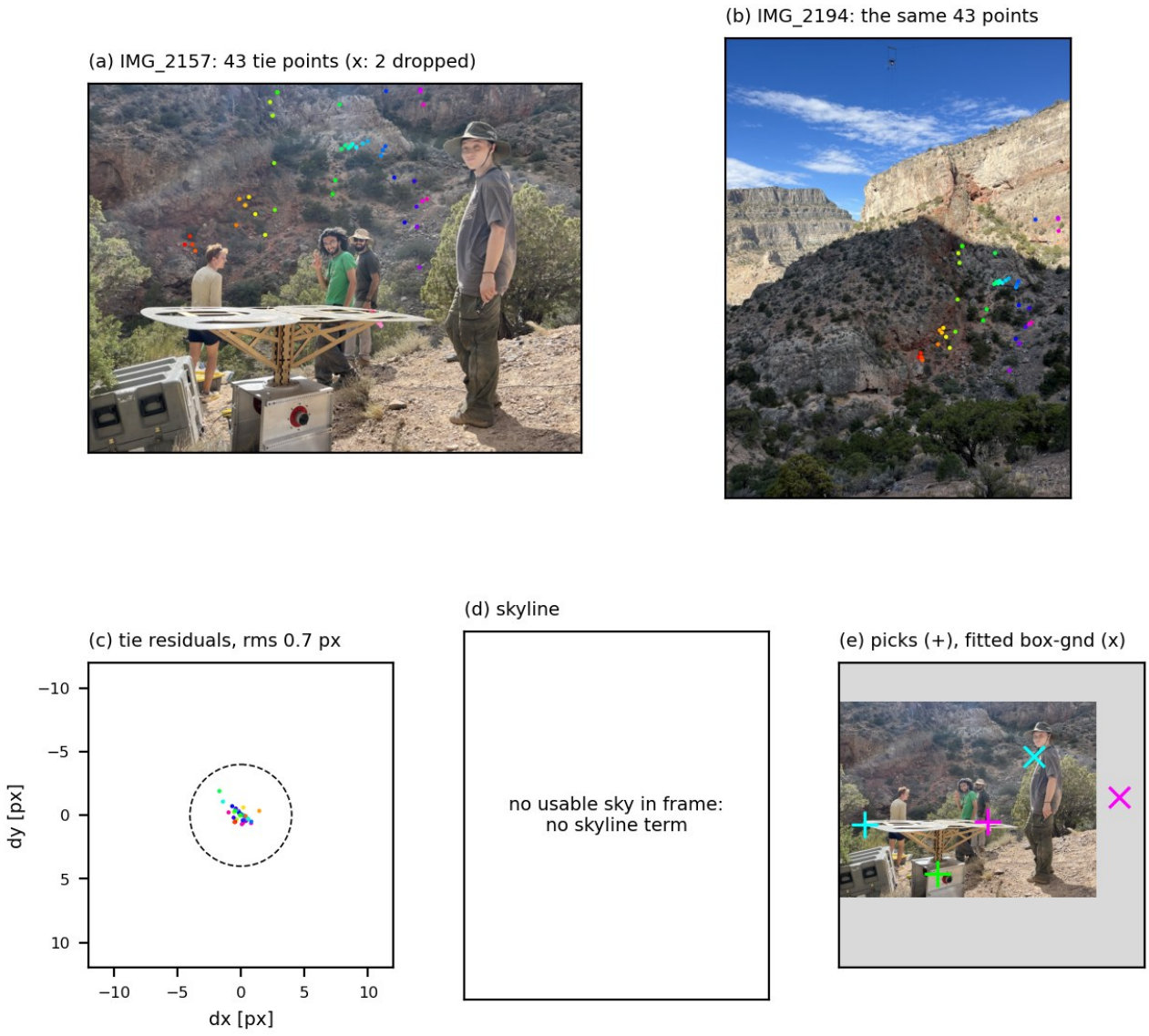


Figure 20: Fit of IMG_2157 (its picks are a check, §9.3). Panels as described in Appendix C; (b) is IMG_2194, the image it shares most tie points with.

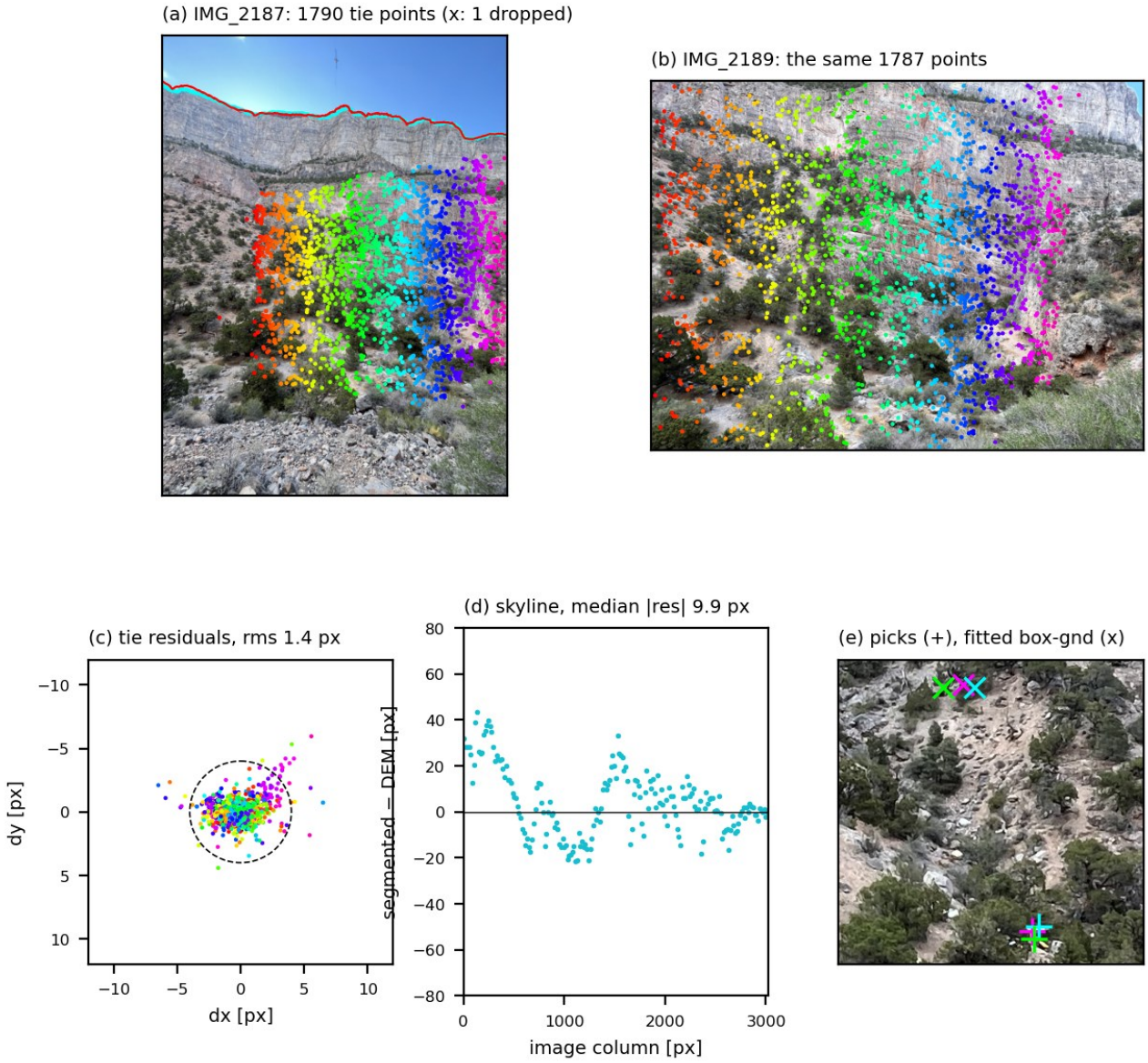


Figure 21: Fit of IMG_2187 (its picks are a check, §9.3). Panels as described in Appendix C; (b) is IMG_2189, the image it shares most tie points with.

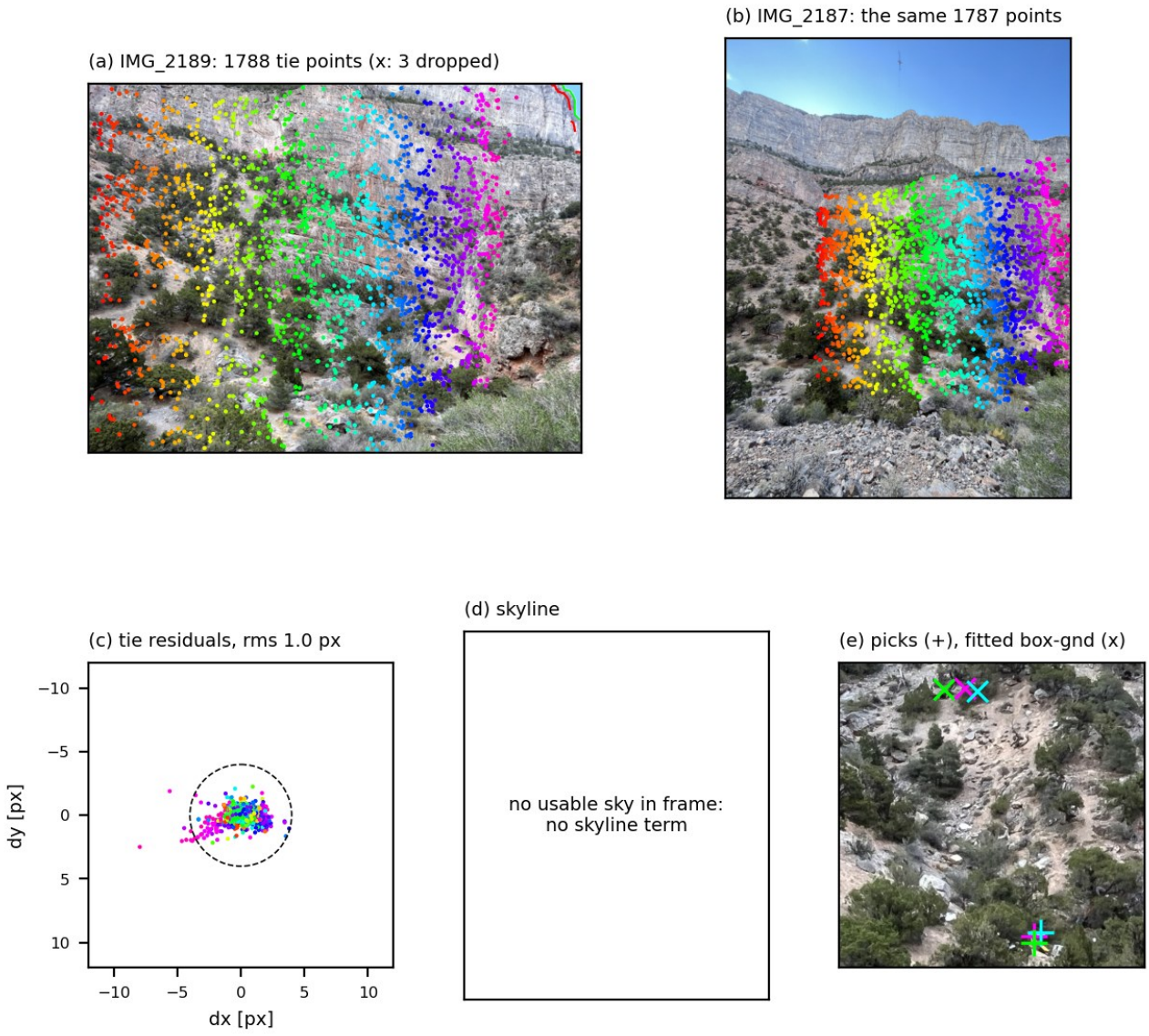


Figure 22: Fit of IMG_2189 (its picks are a check, §9.3). Panels as described in Appendix C; (b) is IMG_2187, the image it shares most tie points with.

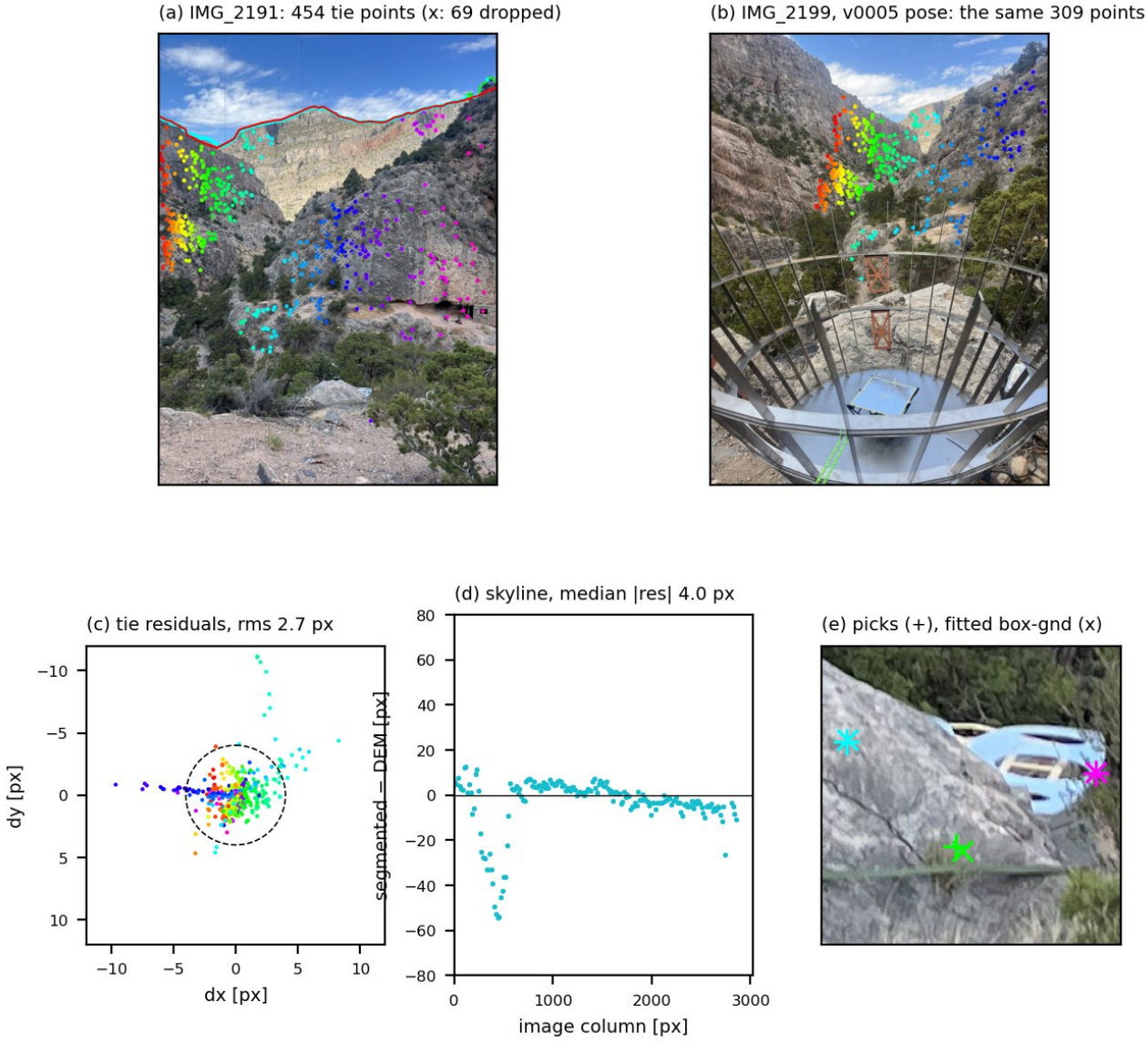


Figure 23: Fit of IMG_2191 (its picks fix `box-gnd`). Panels as described in Appendix C; (b) is IMG_2199, the image it shares most tie points with.

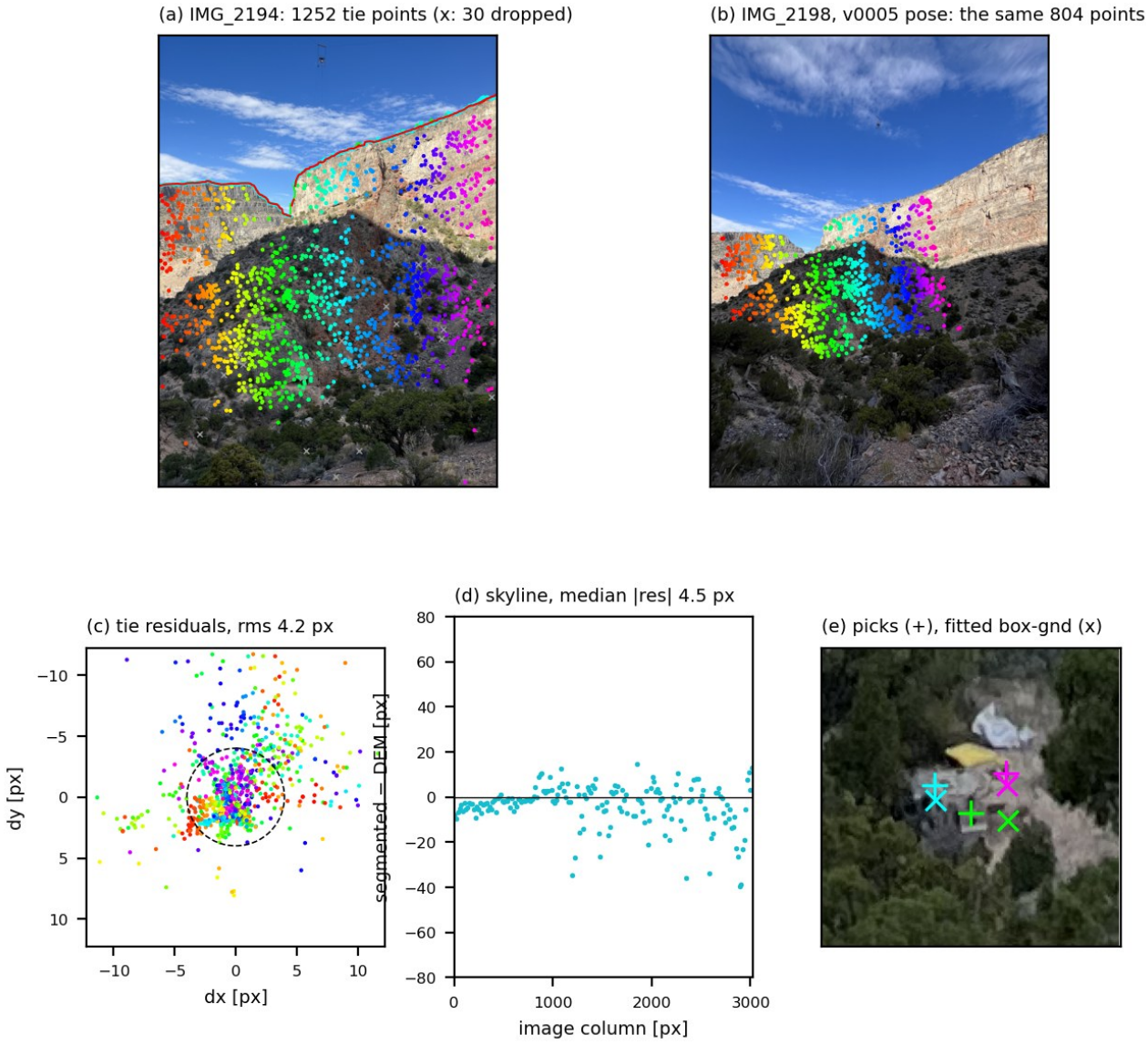


Figure 24: Fit of IMG_2194 (its picks fix `box-gnd`). Panels as described in Appendix C; (b) is IMG_2198, the image it shares most tie points with.

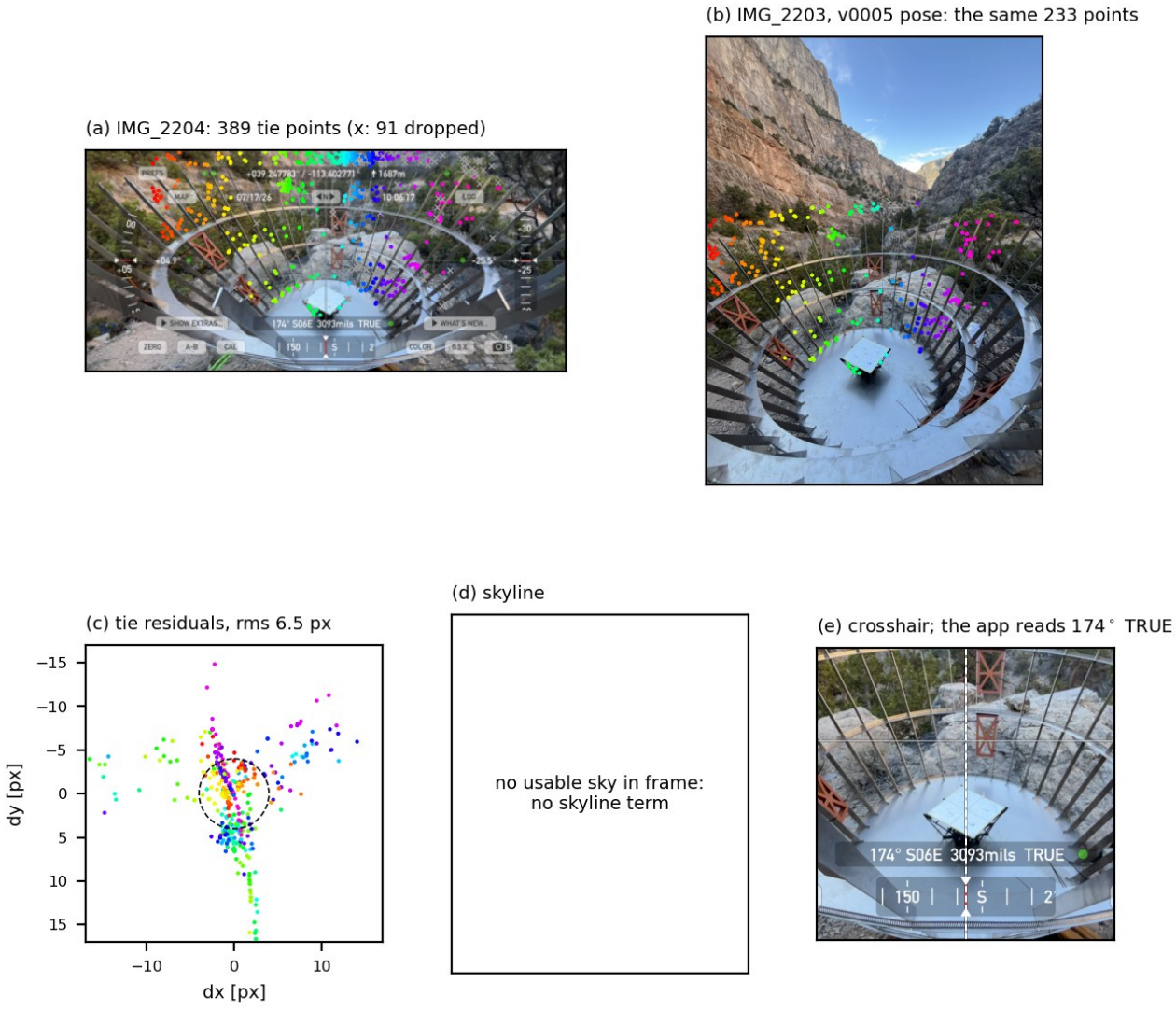


Figure 25: Fit of IMG_2204 (it checks the Theodolite app's readings). Panels as described in Appendix C; (b) is IMG_2203, the image it shares most tie points with.

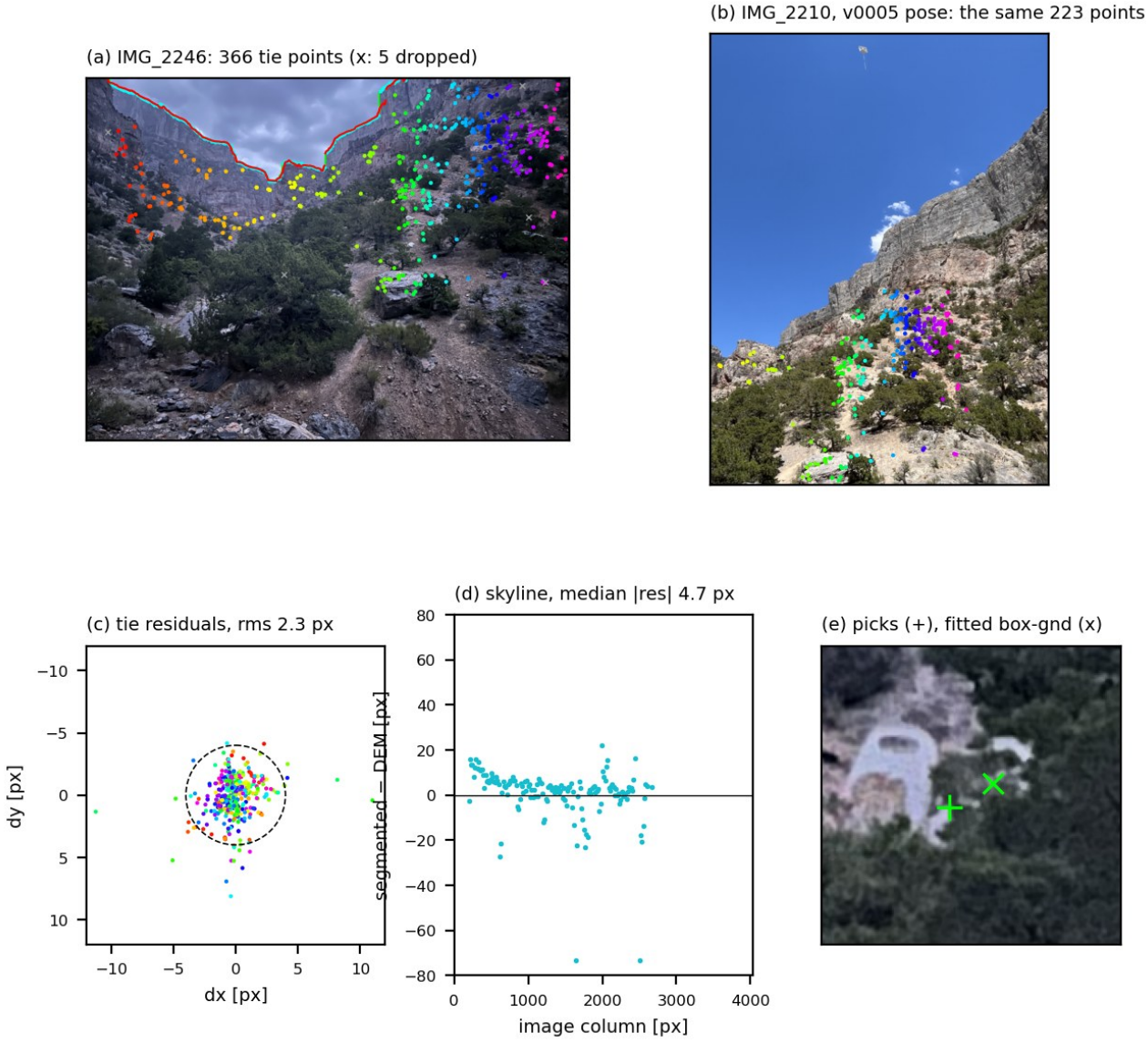


Figure 26: Fit of IMG_2246 (its picks fix `box-gnd`). Panels as described in Appendix C; (b) is IMG_2210, the image it shares most tie points with.

D Reproducing this memo

From this directory, with the project environment:

```
python make_products.py
pdflatex memo-002.tex && pdflatex memo-002.tex
```

Inputs, with the first 16 hex digits of their SHA-256. The DEM, the geometry, the LIDAR constraint and the transmitter-feed orientation are products under `marjum-2026-07/derived/`; each has a manifest recording its code and inputs. The v0005 model inputs (labels, distortion and tie-point observations) are read from the product's copy in `derived/geometry_posterior/v0005/inputs/model_v0005`.

input	SHA-256 (first 16)
marjum-2026-07/imgs/fits/v0005_marjum_geometry/cameras.jsonl	fbbc074ea98312e3
marjum-2026-07/imgs/fits/v0005_marjum_geometry/shared.json	2ec618bff168c1b0
marjum-2026-07/imgs/fits/v0005_marjum_geometry/manifest.json	7028df7d56f01934
marjum-2026-07/imgs/fits/v0001_marjum_geometry/cameras.jsonl	2ed744f833f689e9
marjum-2026-07/imgs/fits/v0001_marjum_geometry/shared.json	234c1d5ab7142318
marjum-2026-07/derived/geometry_memo_inputs/v0001/pilot_review.json	18ac3ba300391bda
marjum-2026-07/derived/geometry_memo_inputs/v0001/manifest.json	758f80c9fc7e2f06
marjum-2026-07/derived/geometry_posterior/v0003/manifest.json	605a8d216491d23f
marjum-2026-07/derived/geometry_posterior/v0003/full_20261006/convergence_report.json	4f9762fbe87cd130
marjum-2026-07/derived/geometry_posterior/v0003/provisional_geometry.npz	06024c369cde40dc
marjum-2026-07/derived/geometry_posterior/v0004/manifest.json	29a747c8b9fc5ce9
marjum-2026-07/derived/geometry_posterior/v0004/run/convergence_report.json	fba0fbdee321e1cd
marjum-2026-07/derived/geometry_posterior/v0004/provisional_geometry.npz	6e611506298f05aa
marjum-2026-07/derived/geometry_posterior/v0005/manifest.json	822fdef2ac96da08
marjum-2026-07/derived/geometry_posterior/v0005/chain_diagnostics.json	5772d8010eef7fd7
marjum-2026-07/derived/geometry_posterior/v0005/posterior_samples.npz	3d3cd0308d0921df
marjum-2026-07/derived/geometry_posterior/v0005/provisional_geometry.npz	e005f1414f976a11
marjum-2026-07/derived/geometry_posterior/v0005/inputs/model_v0005/fit_transmitter.npz	d90338cbc30d506d
marjum-2026-07/derived/geometry_posterior/v0005/inputs/model_v0005/meta.json	4c0e5de2e0f07d0f
marjum-2026-07/derived/dem/v0002/marjum_dem.npz	e890231c970be4ed
marjum-2026-07/derived/lidar_constraint/v0003/results.npz	d28629f3b5c0d5e3
marjum-2026-07/derived/lidar_constraint/v0003/summary.json	30a90a505c043ed0
marjum-2026-07/derived/lidar_constraint/v0003/manifest.json	5b2f5f98d105a2b6
marjum-2026-07/derived/geometry_checks/v0002/manifest.json	19bddb2074c407b9
marjum-2026-07/derived/tx_feed_orientation/v0003/summary.json	3226392671618ba8
marjum-2026-07/derived/tx_feed_orientation/v0003/draw_diagnostics.json	75b1c11e0a90b7f3
marjum-2026-07/derived/tx_feed_orientation/v0003/manifest.json	1b6170e3cd65a60a
marjum-2026-07/derived/tx_feed_orientation/v0001/inputs/tx_feed_picks.json	d5f3fc6a5b4891b3
eigsep (memo, campaign) commit	28e955a
legacy analysis commit	8a09806
generated	2026-10-26 21:58 UTC